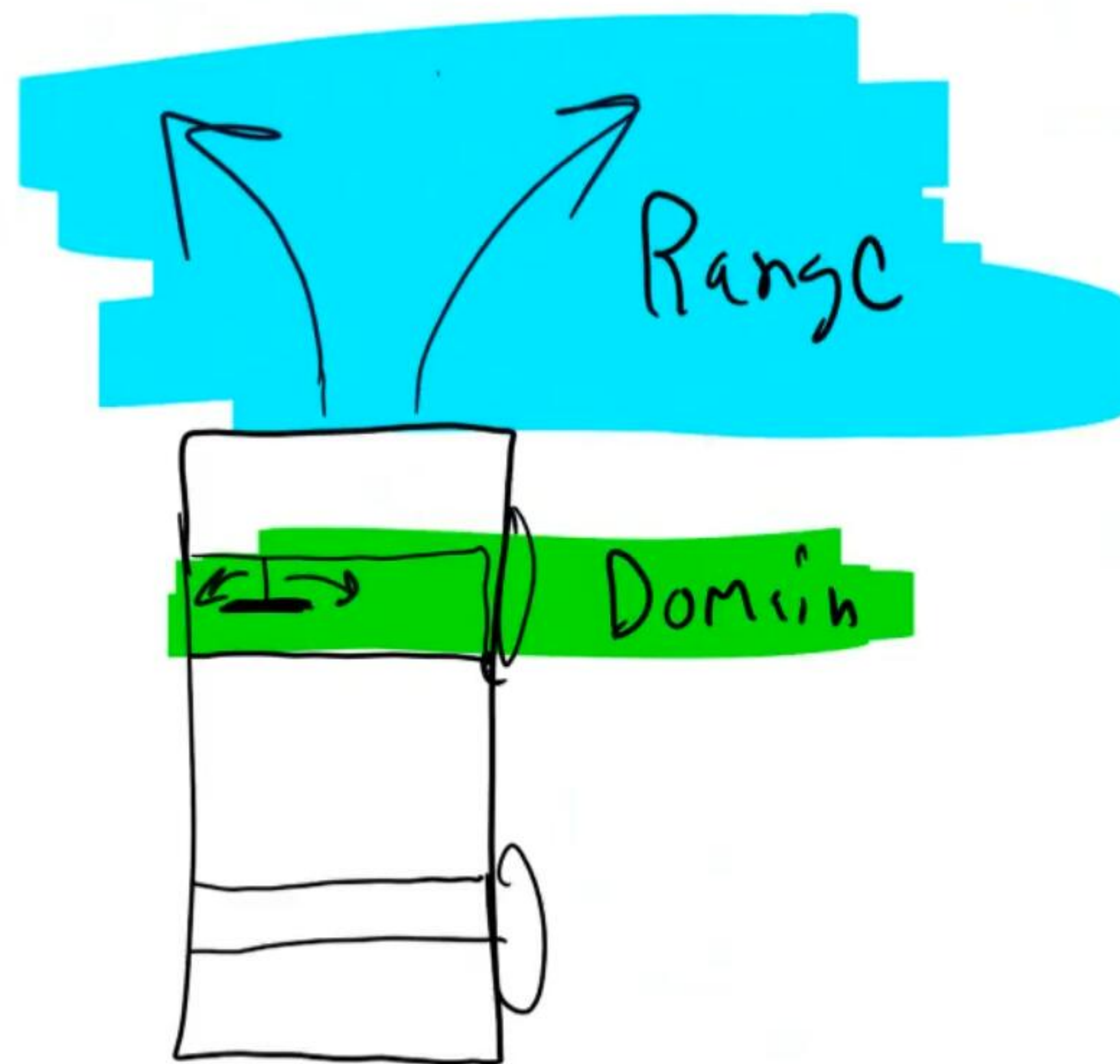
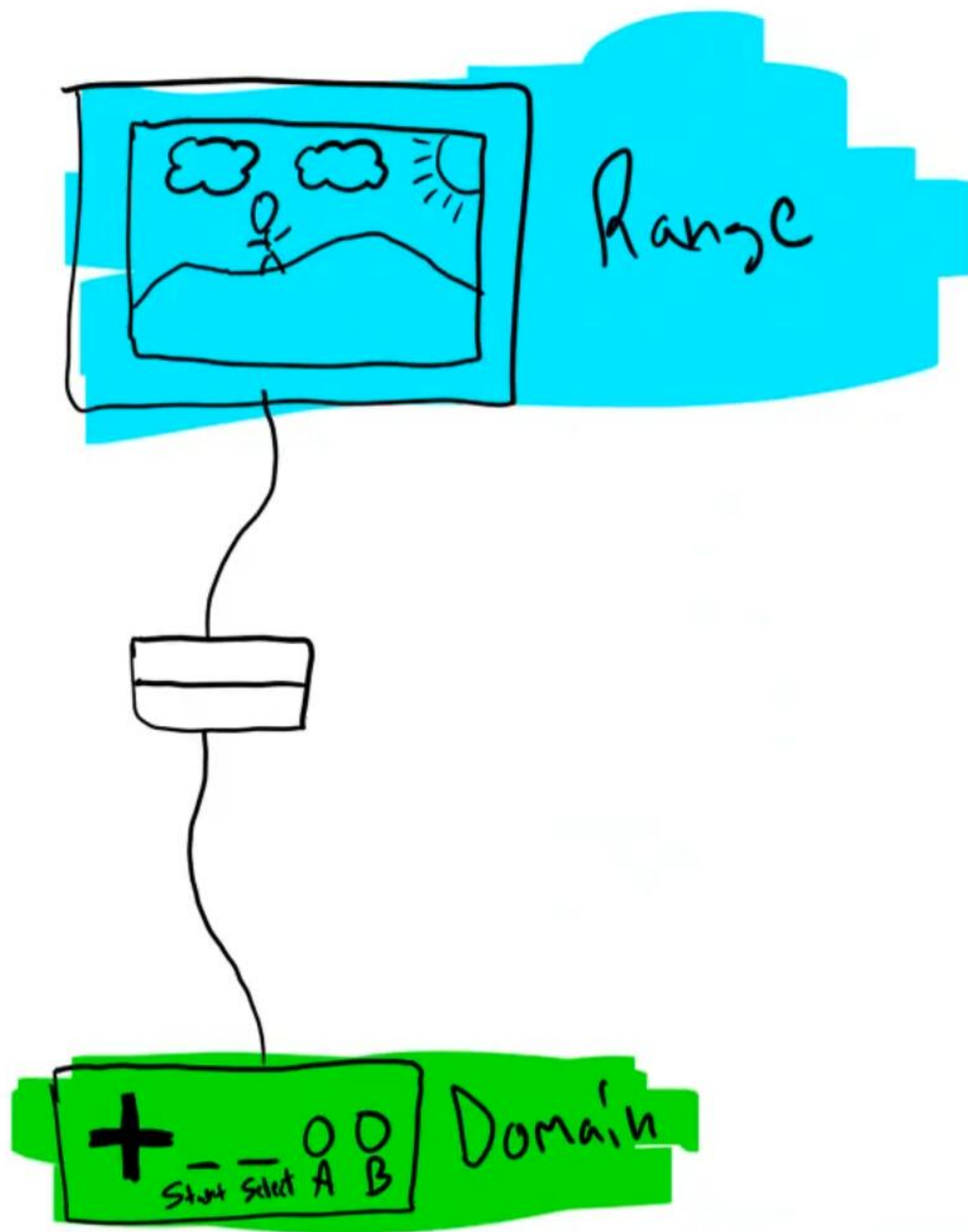




Differentiability and Review on Domain and Interval Notation

Review on the Domain and Range of a Function

Domain: All of the inputs of a function (typically called x-values) that are able to produce outputs (typically called y-values).

Range: All of the outputs of a function that are produced from the inputs in the domain of the function.

★ Imaginary numbers are not allowed to be inputs or outputs of functions.

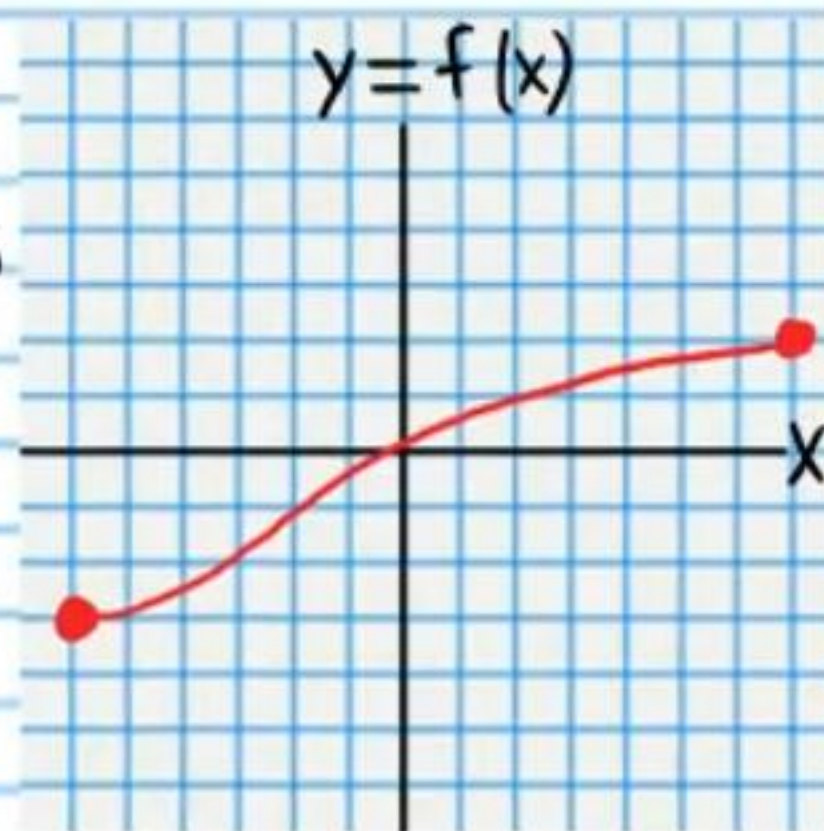
The graphs of functions are given. Use these graphs to write the domain and range of each function in words, set notation, and interval notation.

① Domain

Words: All real numbers between -6 and 7, inclusive.

Set Notation: $\{x \mid -6 \leq x \leq 7\}$

Interval Notation: $[-6, 7]$



Range

Words: All real numbers between -3 and 2, inclusive.

Set Notation: $\{y \mid -3 \leq y \leq 2\}$

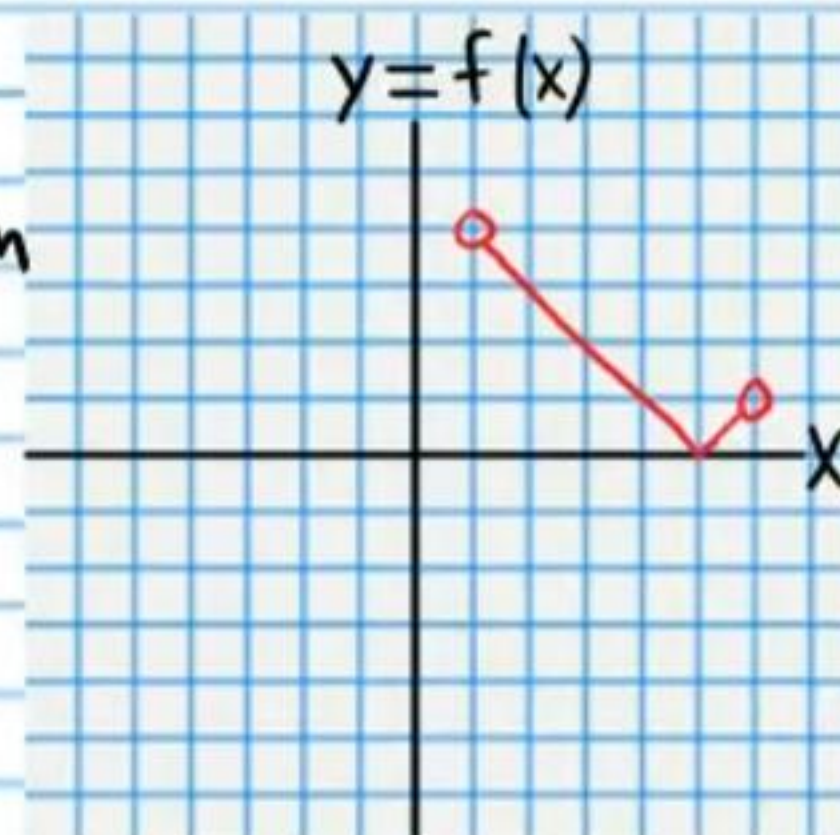
Interval Notation: $[-3, 2]$

② Domain

Words: All real numbers between 1 and 6.

Set Notation: $\{x \mid 1 < x < 6\}$

Interval Notation: $(1, 6)$



Range

Words: All real numbers between zero and 4, including zero.

Set Notation: $\{y \mid 0 \leq y < 4\}$

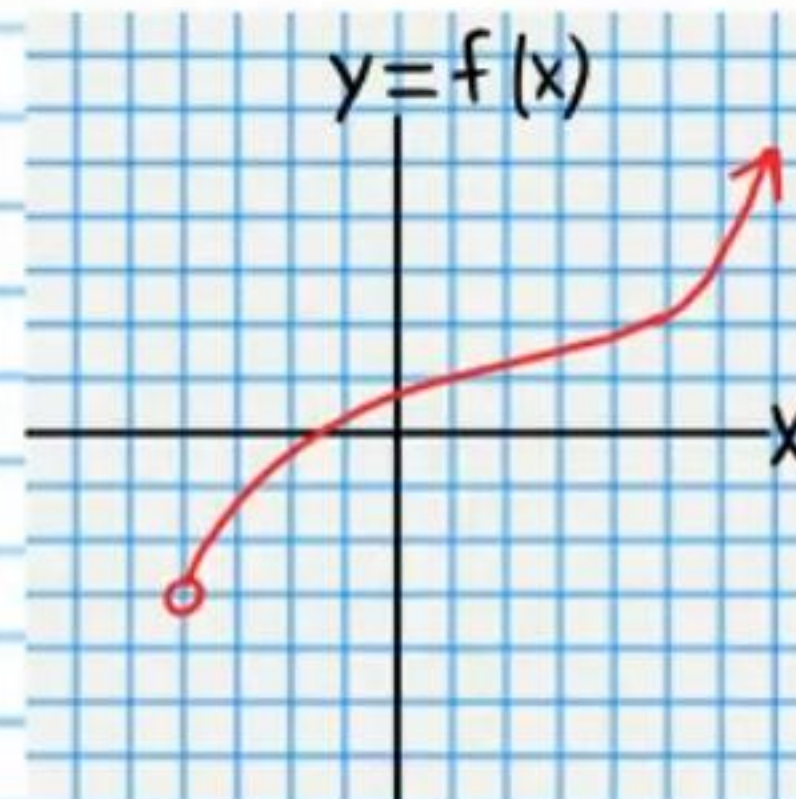
Interval Notation: $[0, 4)$

③ Domain

Words: All real numbers greater than -4.

Set Notation: $\{x \mid x > -4\}$

Interval Notation: $(-4, \infty)$



Range

Words: All real numbers greater than -3.

Set Notation: $\{y \mid y > -3\}$

Interval Notation: $(-3, \infty)$

④ Domain

Words: All real numbers.

Set Notation: All real numbers.

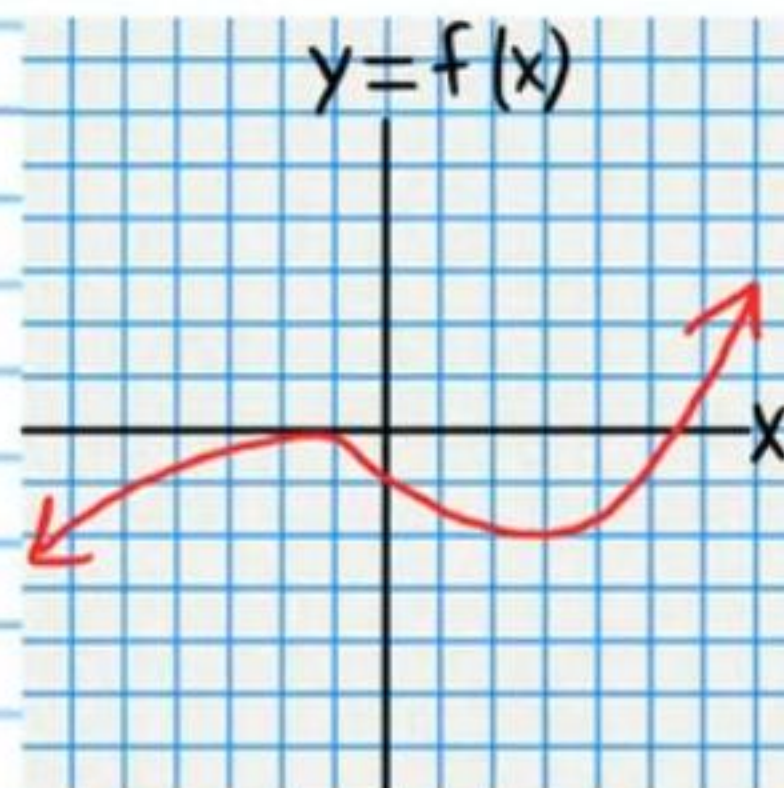
Interval Notation: $(-\infty, \infty)$

Range

Words: All real numbers.

Set Notation: All real numbers

Interval Notation: $(-\infty, \infty)$



⑤ Domain

Words: All real numbers less than -3 or greater than 1.

Set Notation: $\{x \mid x < -3 \text{ or } x > 1\}$

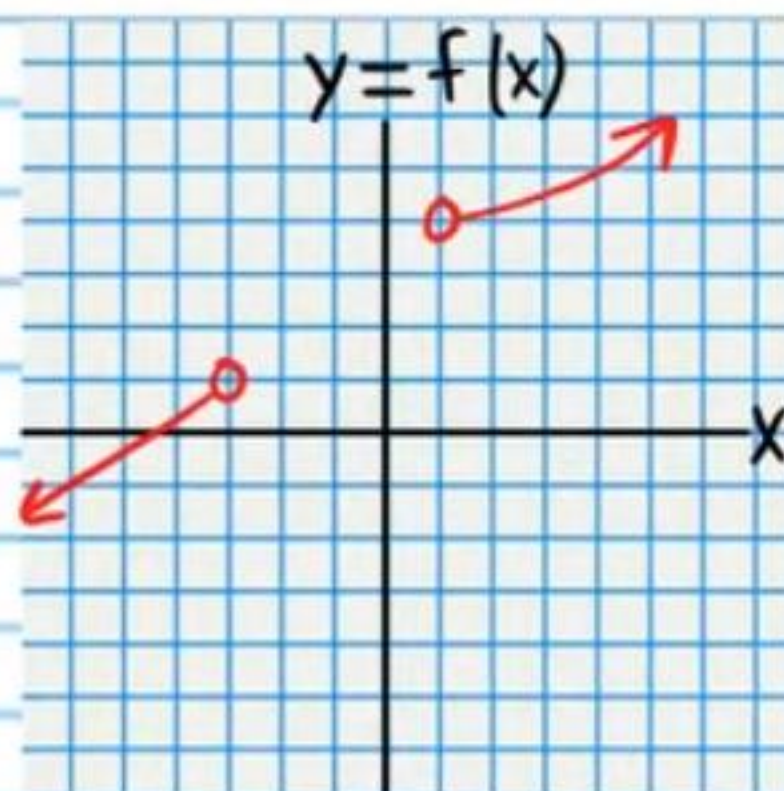
Interval Notation: $(-\infty, -3) \cup (1, \infty)$

Range

Words: All real numbers less than 1 or greater than 4.

Set Notation: $\{y \mid y < 1 \text{ or } y > 4\}$

Interval Notation: $(-\infty, 1) \cup (4, \infty)$



⑥ Domain

Words: All real numbers except 4.

Set Notation: $\{x \mid x \neq 4\}$

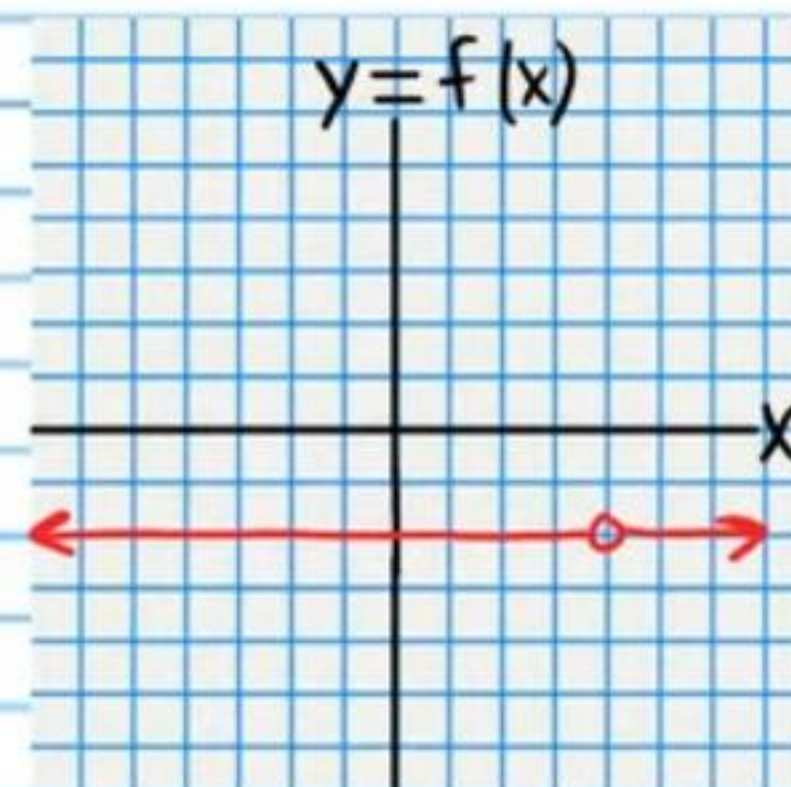
Interval Notation: $(-\infty, 4) \cup (4, \infty)$

Range

Words: The only value in the range is -2.

Set Notation: $\{y \mid y = -2\}$

Interval Notation: $[-2, -2]$



⑦ Domain

Words: All real numbers except 2.

Set Notation: $\{x \mid x \neq 2\}$

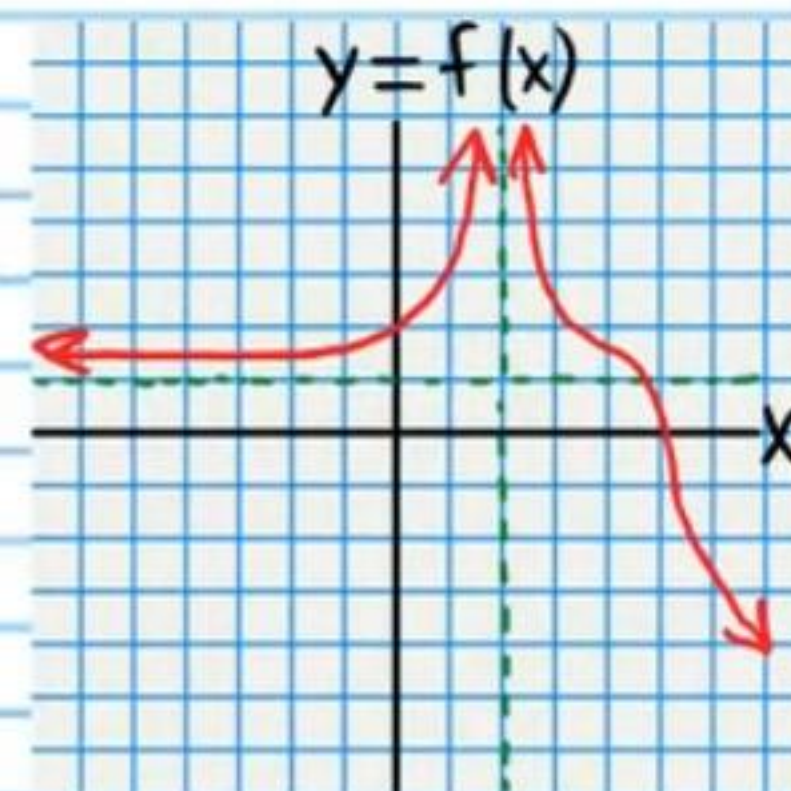
Interval Notation: $(-\infty, 2) \cup (2, \infty)$

Range

Words: All real numbers.

Set Notation: All real numbers.

Interval Notation: $(-\infty, \infty)$



⑧ Domain

Words: All real numbers.

Set Notation: All real numbers.

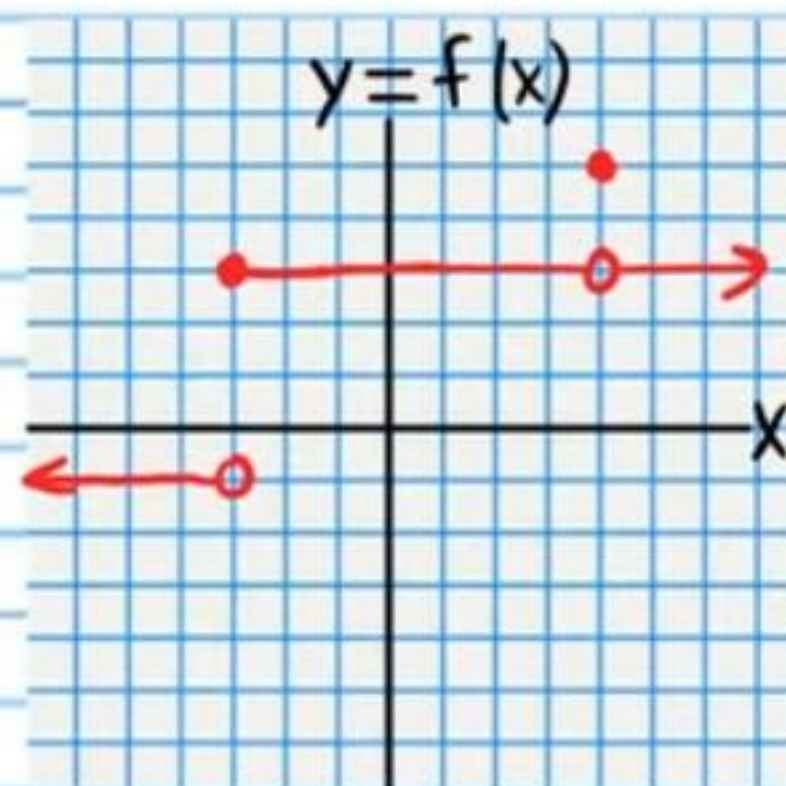
Interval Notation: $(-\infty, \infty)$

Range

Words: The only values in the range are -1 , 3 , and 5 .

Set Notation: $\{y \mid y = -1 \text{ or } y = 3 \text{ or } y = 5\}$

Interval Notation: $[-1, -1] \cup [3, 3] \cup [5, 5]$



● ⑨ Domain

Words: All real numbers between 2 and 5 .

Set Notation: $\{x \mid 2 < x < 5\}$

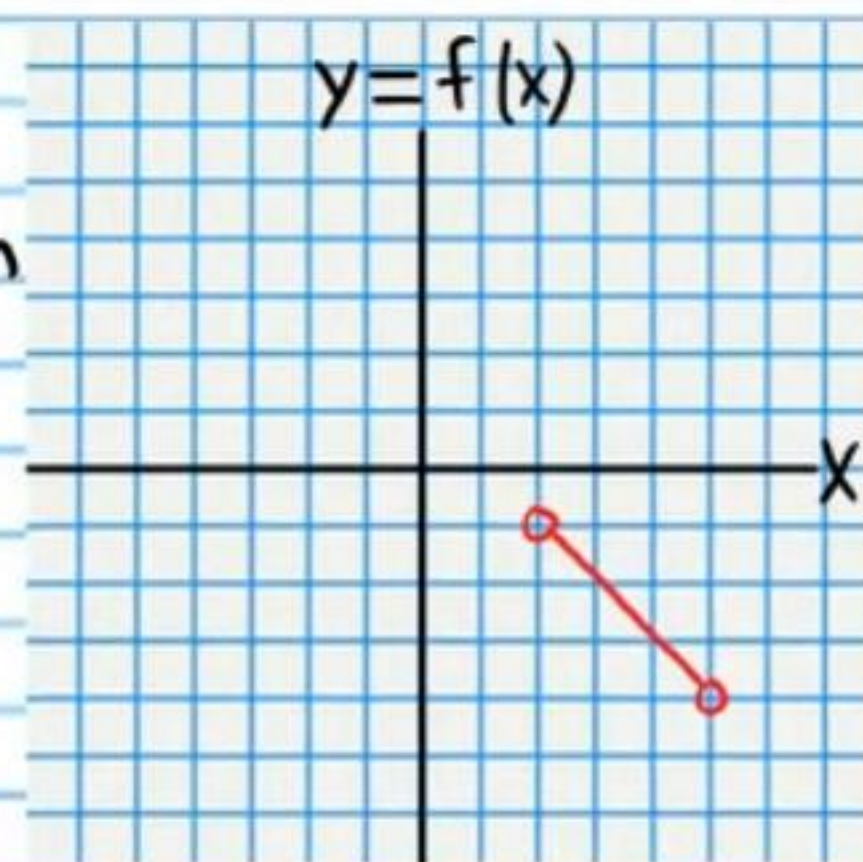
Interval Notation: $(2, 5)$

Range

Words: All real numbers between -4 and -1 .

Set Notation: $\{y \mid -4 < y < -1\}$

Interval Notation: $(-4, -1)$



⑩ Domain

Words: All real numbers between 0 and 5 , including 5 .

Set Notation: $\{x \mid 0 < x \leq 5\}$

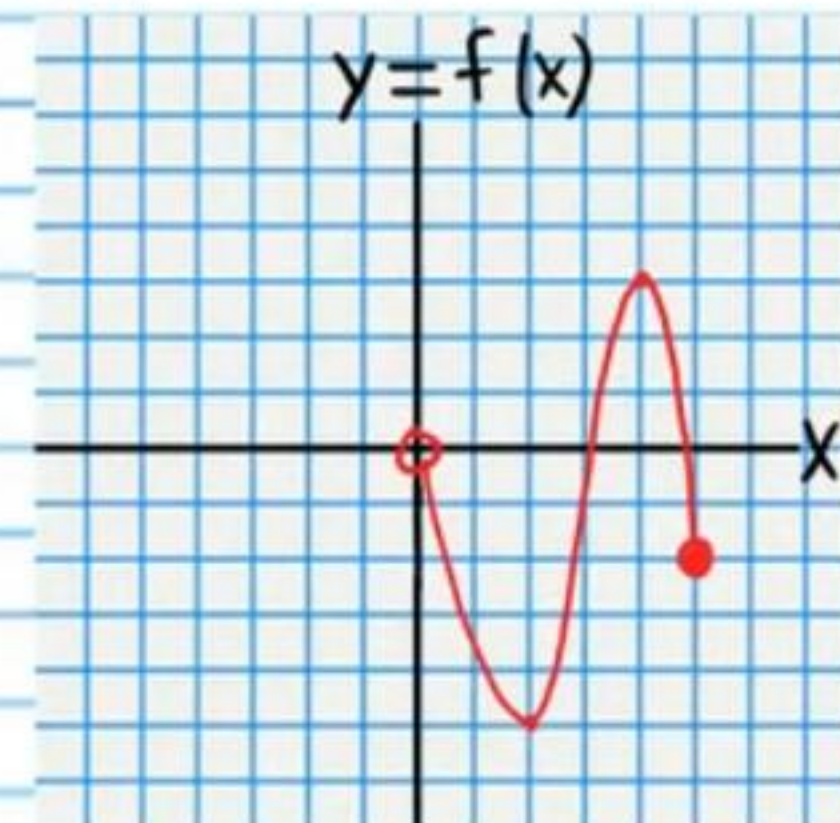
Interval Notation: $(0, 5]$

Range

Words: All real numbers between -5 and 3 , inclusive.

Set Notation: $\{y \mid -5 \leq y \leq 3\}$

Interval Notation: $[-5, 3]$



⑪ Domain

Words: All real numbers less than or equal to 5 .

Set Notation: $\{x \mid x \leq 5\}$

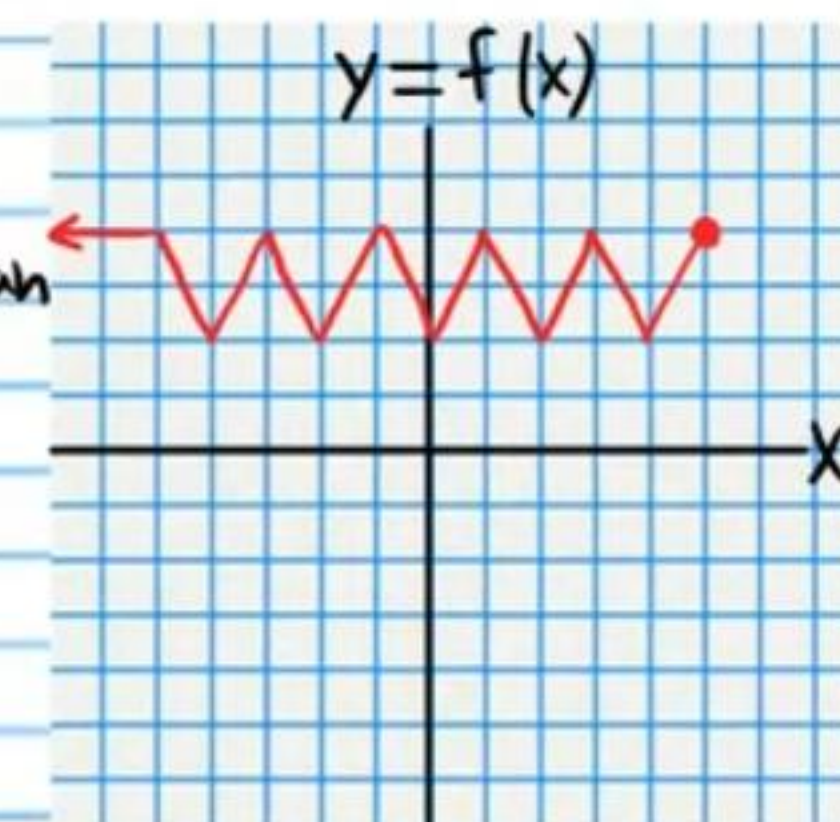
Interval Notation: $(-\infty, 5]$

Range

Words: All real numbers between 2 and 4 , inclusive.

Set Notation: $\{y \mid 2 \leq y \leq 4\}$

Interval Notation: $[2, 4]$



⑫ Domain

Words: All real numbers,

Set Notation: All real numbers,

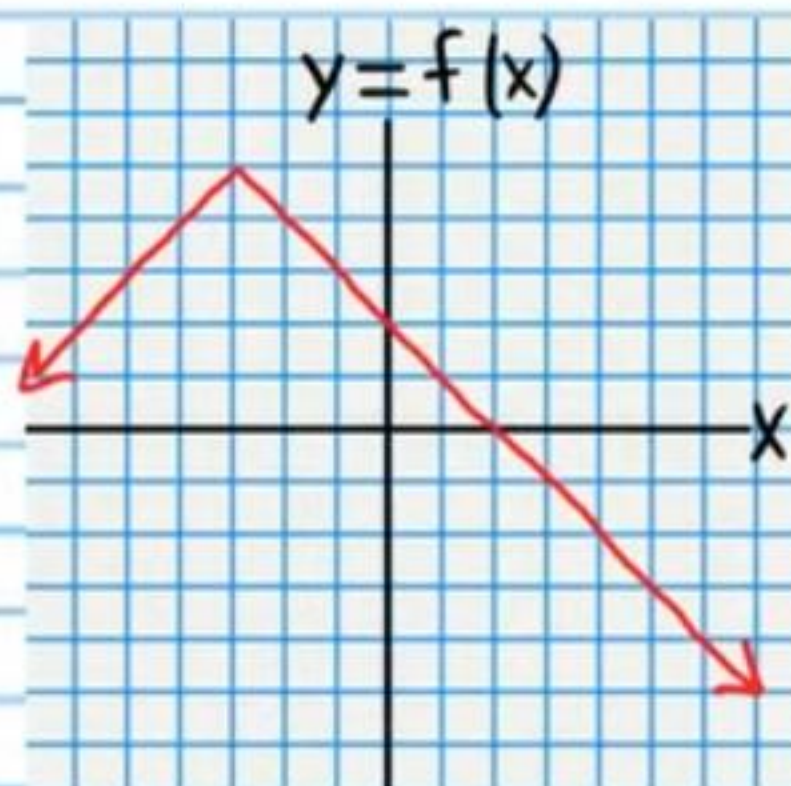
Interval Notation: $(-\infty, \infty)$

Range

Words: All real numbers less than or equal to 5,

Set Notation: $\{y \mid y \leq 5\}$

Interval Notation: $(-\infty, 5]$



⑬ Domain

Words: All real numbers less than 2 or greater than 4.

Set Notation: $\{x \mid x < 2 \text{ or } x > 4\}$

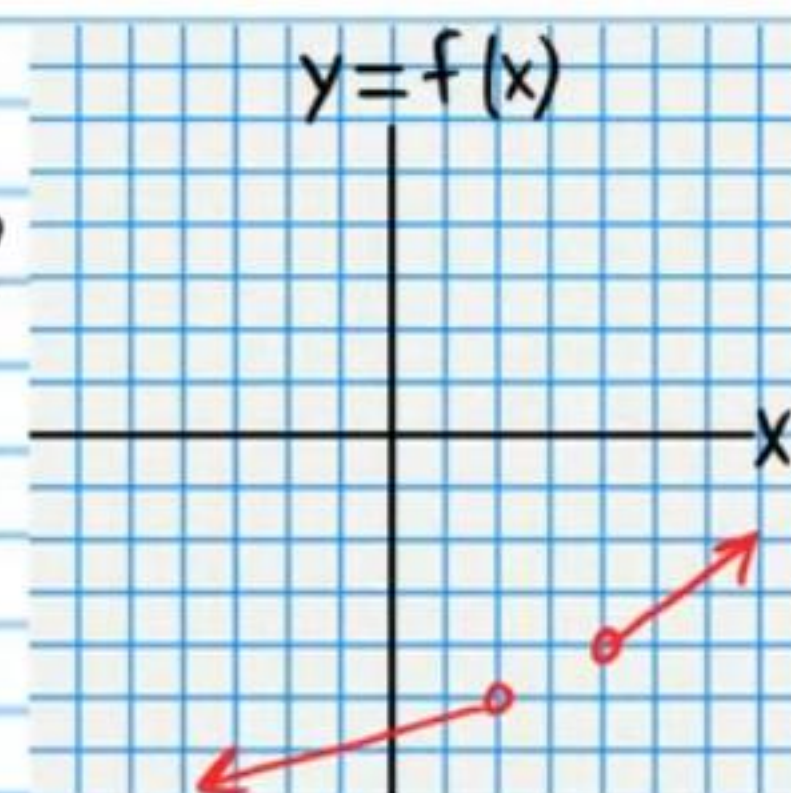
Interval Notation: $(-\infty, 2) \cup (4, \infty)$

Range

Words: All real numbers less than -5 or greater than -4.

Set Notation: $\{y \mid y < -5 \text{ or } y > -4\}$

Interval Notation: $(-\infty, -5) \cup (-4, \infty)$



⑭ Domain

Words: All real numbers except -3 and 1.

Set Notation: $\{x \mid x \neq -3 \text{ and } x \neq 1\}$

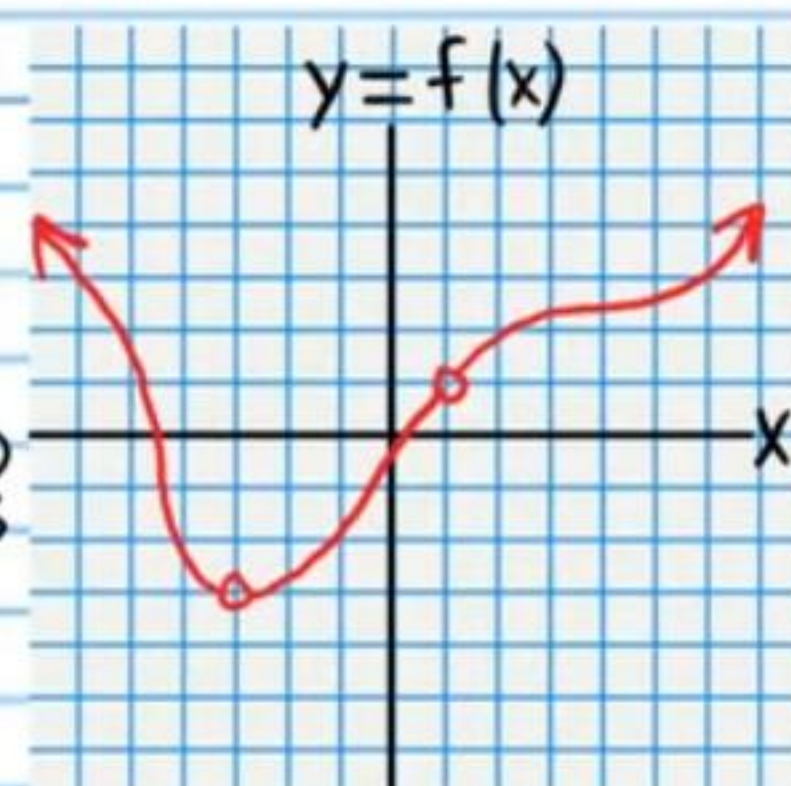
Interval Notation: $(-\infty, -3) \cup (-3, 1) \cup (1, \infty)$

Range

Words: All real numbers greater than -3.

Set Notation: $\{y \mid y > -3\}$

Interval Notation: $(-3, \infty)$



⑮ Domain

Words: All real numbers except -4 and 3.

Set Notation: $\{x \mid x \neq -4 \text{ and } x \neq 3\}$

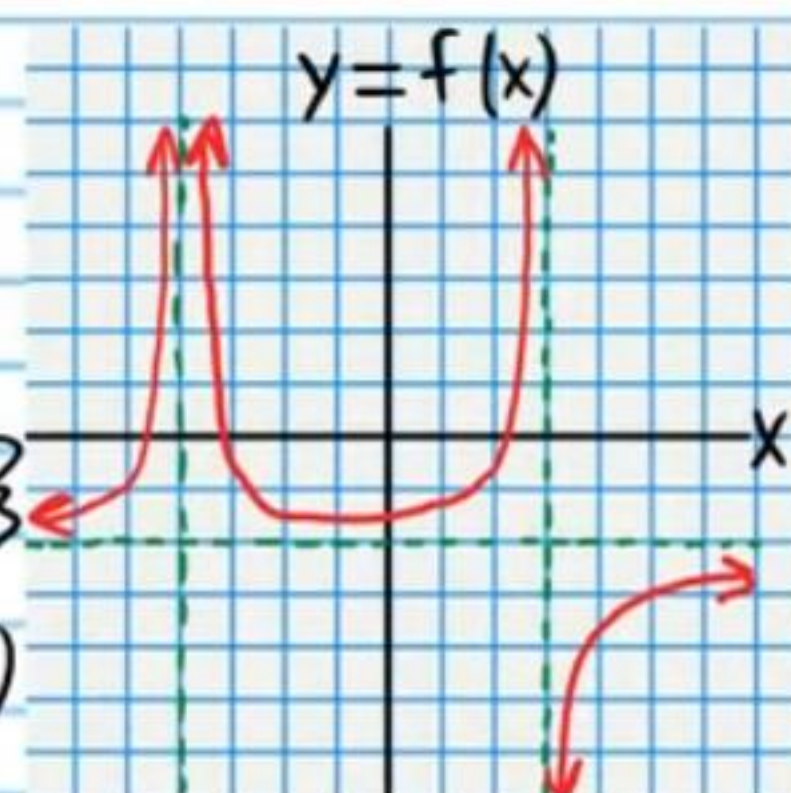
Interval Notation: $(-\infty, -4) \cup (-4, 3) \cup (3, \infty)$

Range

Words: All real numbers except -2.

Set Notation: $\{y \mid y \neq -2\}$

Interval Notation: $(-\infty, -2) \cup (-2, \infty)$



⑩ Domain

Words: All real numbers except -4 .

Set Notation: $\{x \mid x \neq -4\}$

Interval Notation: $(-\infty, -4) \cup (-4, \infty)$

Range

Words: -5 or all real numbers between -3 and 4 , including -3 .

Set Notation: $\{y \mid y = -5 \text{ or } -3 \leq y < 4\}$

Interval Notation: $[-5, -5] \cup [-3, 4)$

Find the domain of each function below.

• ⑪ $f(x) = \frac{1}{x+2}$ Domain

Reasoning: A fraction with zero as its denominator is undefined, so $x+2$ cannot be zero.

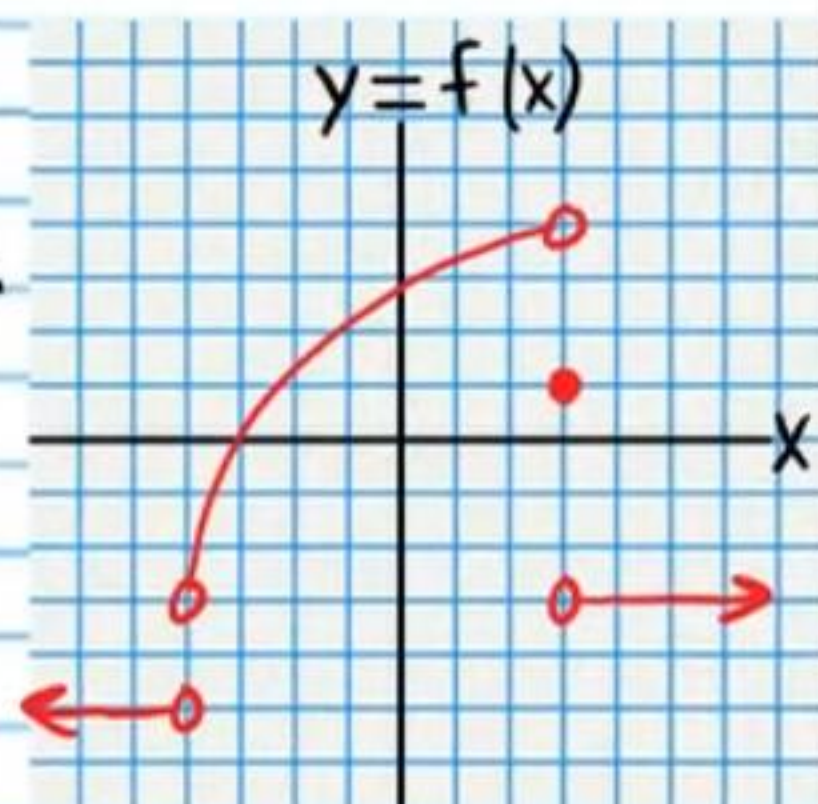
$$x+2 \neq 0$$

$$x \neq -2$$

Words: All real numbers except -2 .

Set Notation: $\{x \mid x \neq -2\}$

Interval Notation: $(-\infty, -2) \cup (-2, \infty)$



⑫ $g(x) = \sqrt{x+5}$

Reasoning: The square root of a negative number is an imaginary number, so $x+5$ must be greater than or equal to zero.

$$x+5 \geq 0$$

$$x \geq -5$$

Domain

Words: All real numbers greater than or equal to -5 .

Set Notation: $\{x \mid x \geq -5\}$

Interval Notation: $[-5, \infty)$

①9 $h(x) = 3 - 4 \log_8(4 - \frac{x}{2})$

Reasoning: Both zero and negative numbers produce an undefined logarithmic expression, so $4 - \frac{x}{2}$ must be greater than zero.

$$4 - \frac{x}{2} > 0$$

$$-\frac{x}{2} > -4$$

$$x < 8$$

Domain

Words: All real numbers less than 8.

Set Notation: $\{x \mid x < 8\}$

Interval Notation: $(-\infty, 8)$

● ②0 $g(x) = \frac{-7}{x-4}$

Reasoning: A fraction with zero as its denominator is undefined, so $x-4$ cannot be zero.

$$x - 4 \neq 0$$

$$x \neq 4$$

Domain

Words: All real numbers except 4.

Set Notation: $\{x \mid x \neq 4\}$

Interval Notation: $(-\infty, 4) \cup (4, \infty)$

②1 $h(x) = \sqrt{-2x+1}$

Reasoning: The square root of a negative number is an imaginary number, so $-2x+1$ must be greater than or equal to zero.

$$-2x + 1 \geq 0$$

$$-2x \geq -1$$

$$x \leq \frac{1}{2}$$

Domain

Words: All real numbers less than or equal to $\frac{1}{2}$.

Set Notation: $\{x \mid x \leq \frac{1}{2}\}$

Interval Notation: $(-\infty, \frac{1}{2}]$

②② $f(x) = \log_3 X$

Reasoning: Both zero and negative numbers produce an undefined logarithmic expression, so X must be greater than zero.

$$X > 0$$

Domain

Words: All real numbers greater than zero.

Set Notation: $\{x \mid x > 0\}$

Interval Notation: $(0, \infty)$

● ②③ $f(x) = |x|$

Reasoning: The function has no square roots, fractions, logarithms, or other things that might restrict its domain.

Domain

Words: All real numbers.

Set Notation: All real numbers.

Interval Notation: $(-\infty, \infty)$

②④ $h(x) = x^3$

Reasoning: The function has no square roots, fractions, logarithms, or other things that might restrict its domain.

Domain

Words: All real numbers.

Set Notation: All real numbers.

Interval Notation: $(-\infty, \infty)$

● ②⑤ $g(x) = 2^{x+3}$

Reasoning: The function has no square roots, fractions, logarithms, or other things that might restrict its domain.

Domain

Words: All real numbers.

Set Notation: All real numbers.

Interval Notation: $(-\infty, \infty)$

②⑥ $f(x) = 3x + 6$

Reasoning: The function has no square roots, fractions, logarithms, or other things that might restrict its domain.

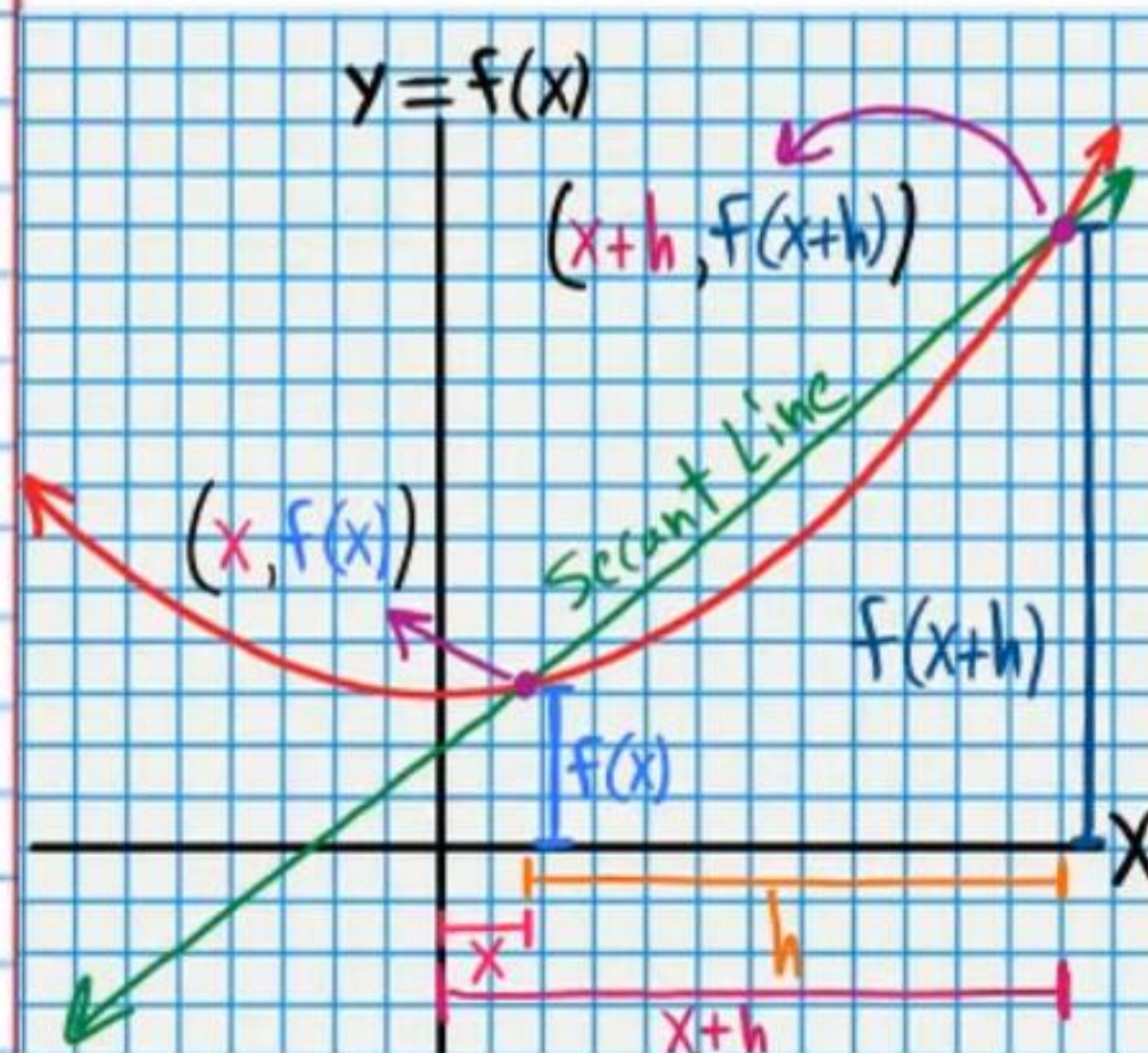
Domain

Words: All real numbers.

Set Notation: All real numbers,

Interval Notation: $(-\infty, \infty)$

Review on the Definition of the Derivative



The Slope of a Line Given Points (x_1, y_1) and (x_2, y_2) on the Line.

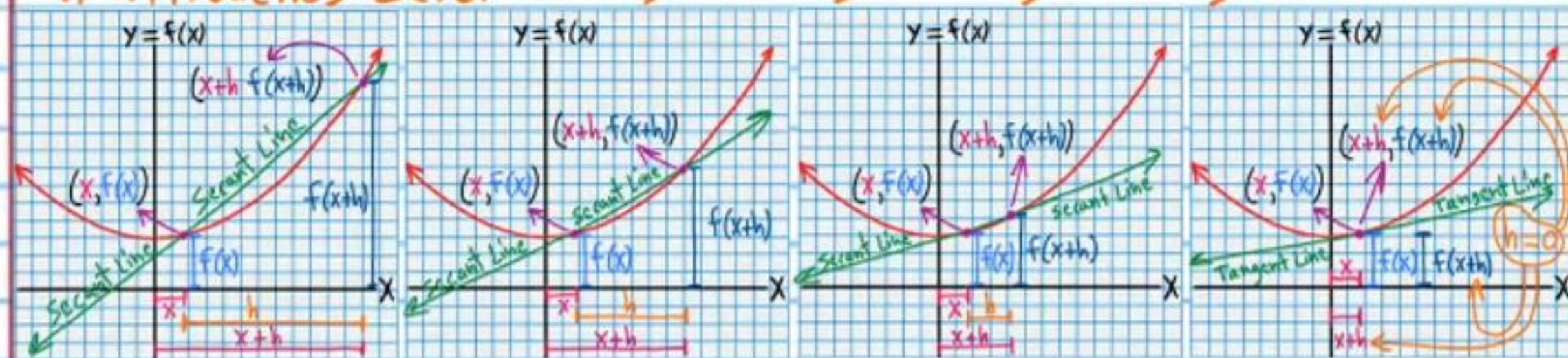
$$M = \frac{y_2 - y_1}{x_2 - x_1}$$

The Slope of the Secant Line Given Points $(x, f(x))$ and $(x+h, f(x+h))$ on the Line.

$$M_{\text{sec}} = \frac{f(x+h) - f(x)}{x+h - x}$$

$$M_{\text{sec}} = \text{Average rate of change of the function on the interval } (x, x+h) = \frac{f(x+h) - f(x)}{h}$$

"h" approaches zero. $\longrightarrow$ (The Difference Quotient)



The Slope of the Tangent Line

$$M_{\text{tan}} = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

(the instantaneous rate of change of the function at "x.")

The Derivative of $f(x)$

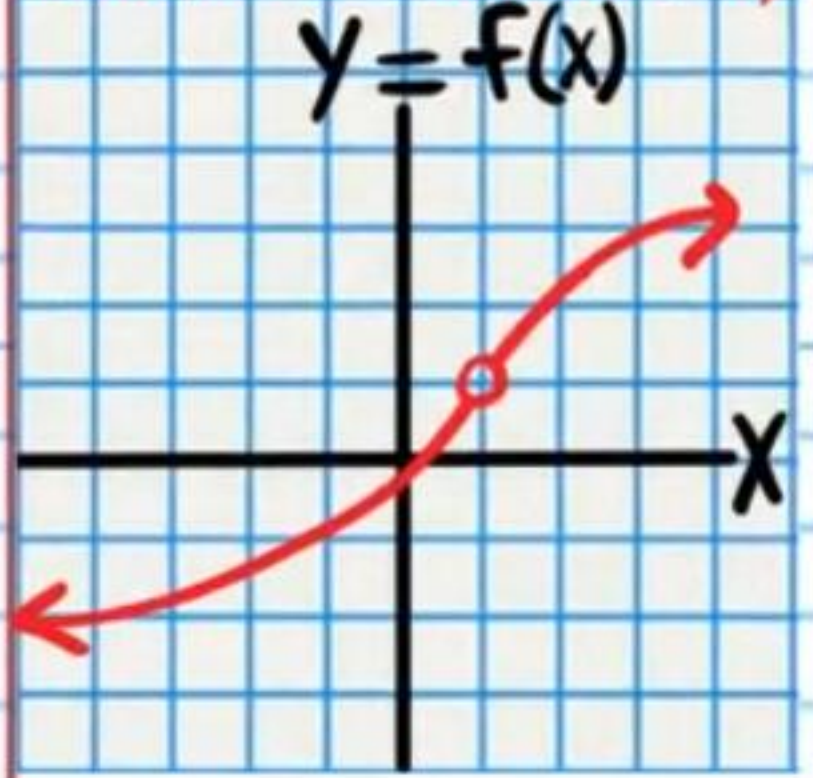
$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Review on Continuity

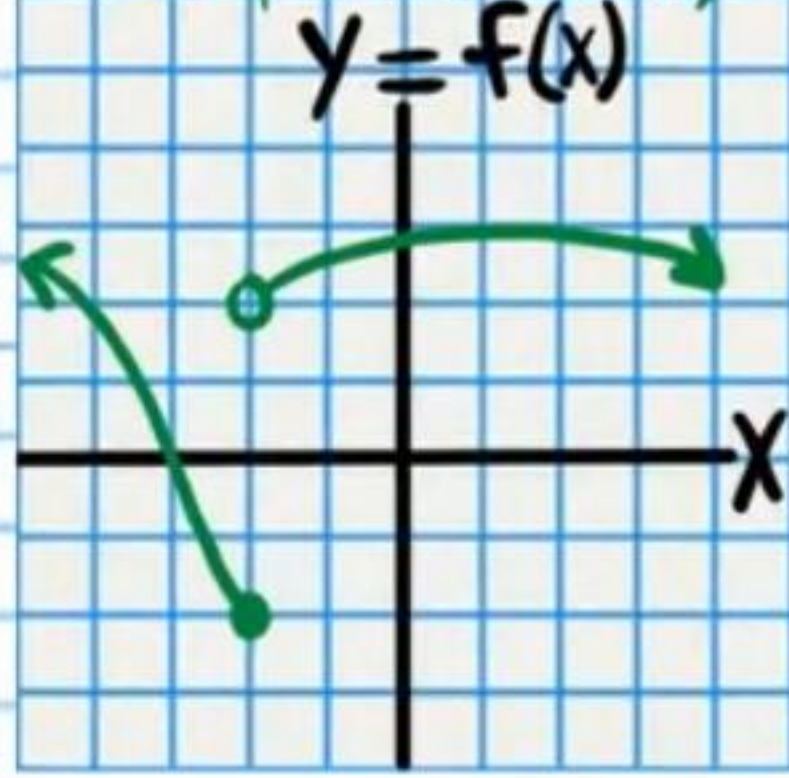
Continuous Function: A function whose graph has no breaks (i.e. holes, jumps, or vertical asymptotes).

Examples of Discontinuity

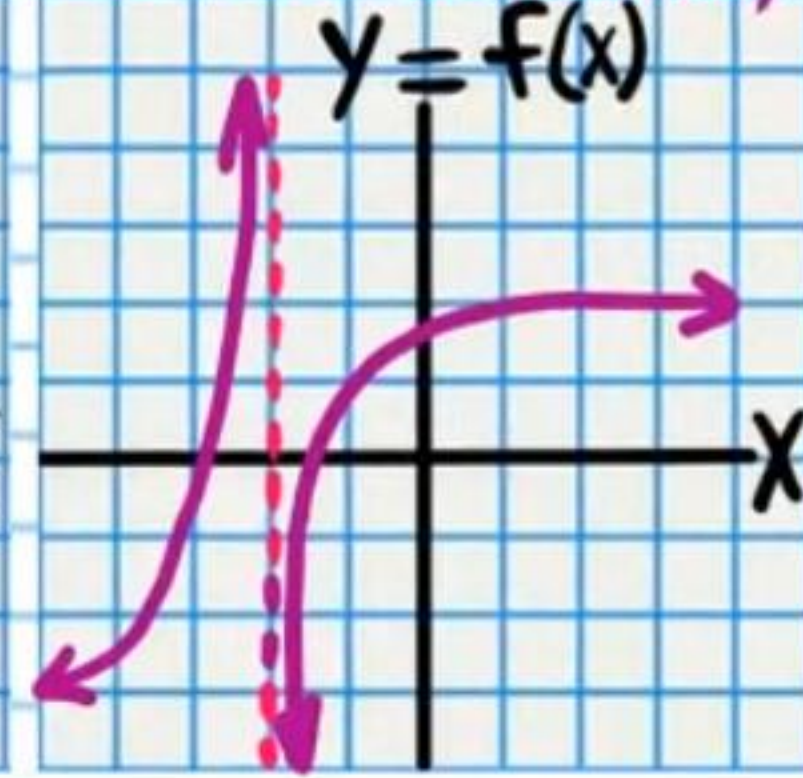
Removable Discontinuity



Jump Discontinuity



Infinite Discontinuity



The Definition of Continuity at Some X-value: A function is continuous at $x=a$ if $\lim_{x \rightarrow a} f(x) = f(a)$.

In order for $\lim_{x \rightarrow a} f(x) = f(a)$, three things must be true.

i) $f(a)$ must exist.

ii) $\lim_{x \rightarrow a} f(x)$ must exist (implying that $\lim_{x \rightarrow a^+} f(x)$ and $\lim_{x \rightarrow a^-} f(x)$ both exist and are equal).

iii) $f(a)$ and $\lim_{x \rightarrow a} f(x)$ must be equal.

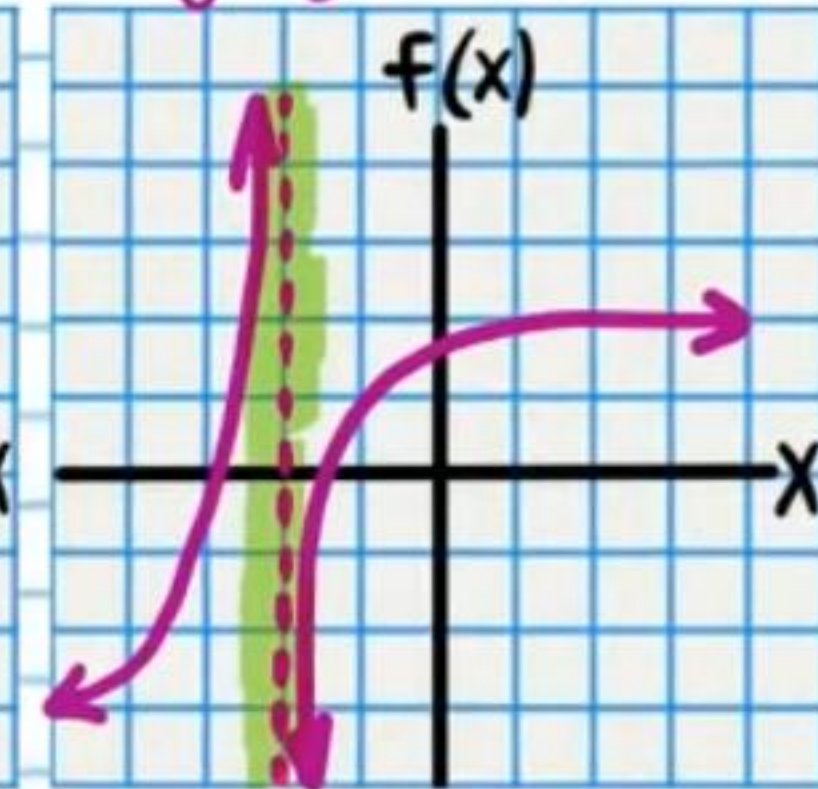
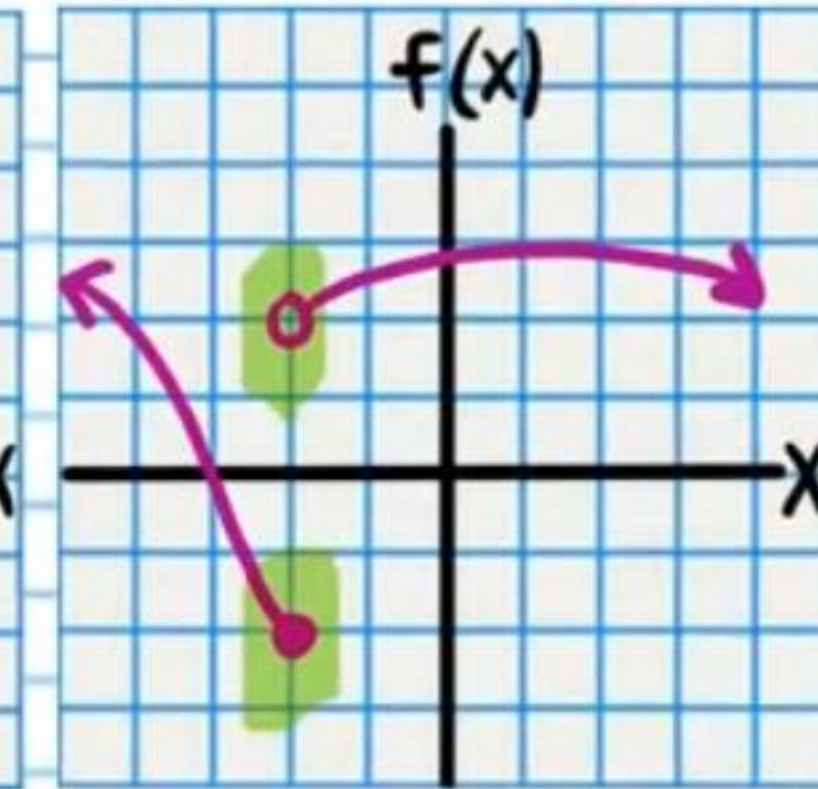
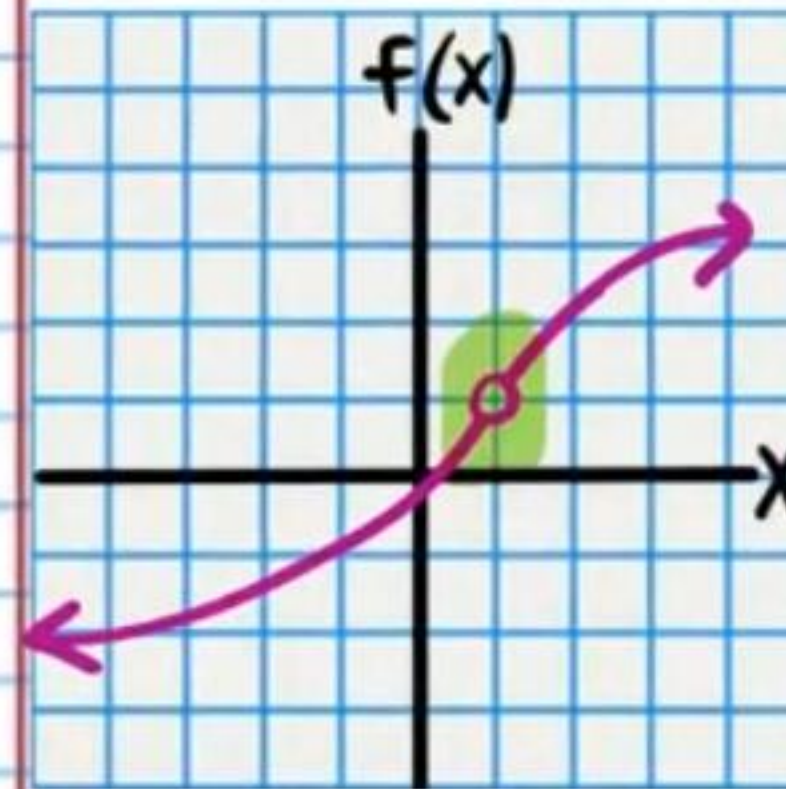
Differentiability

Differentiability: A function is differentiable at $x=a$ if $f'(a)$ exists.

Scenarios Where $f'(a)$ Does Not Exist

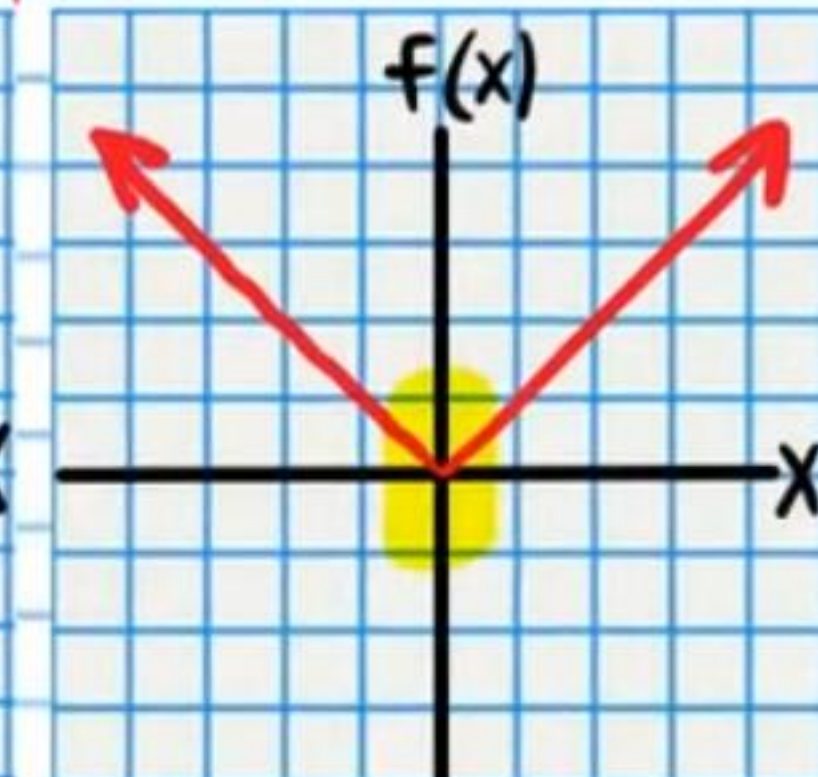
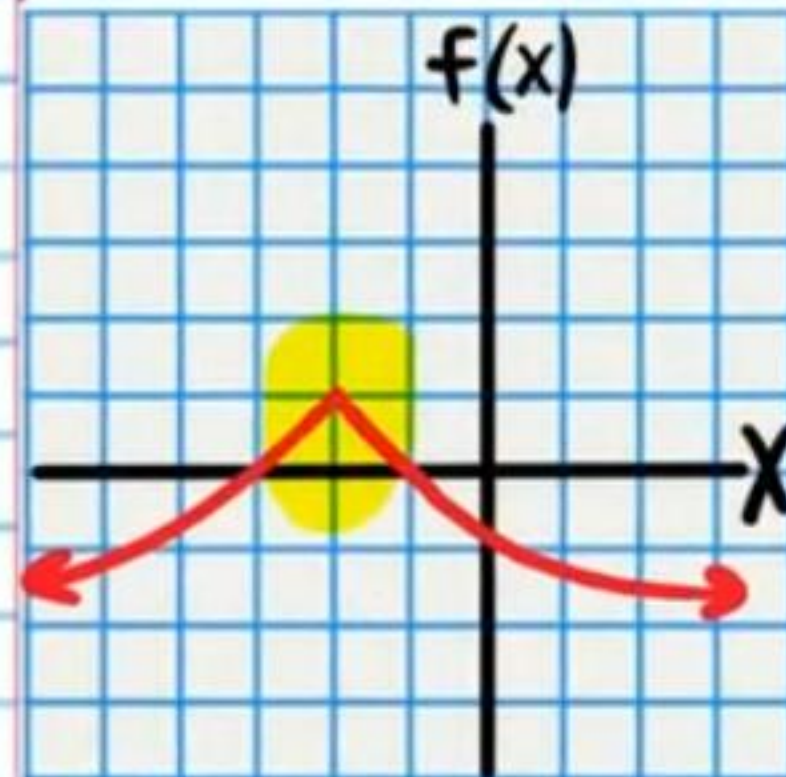
If a function is discontinuous at $x=a$, then it is not differentiable at $x=a$.

These functions are not differentiable at each highlighted location.

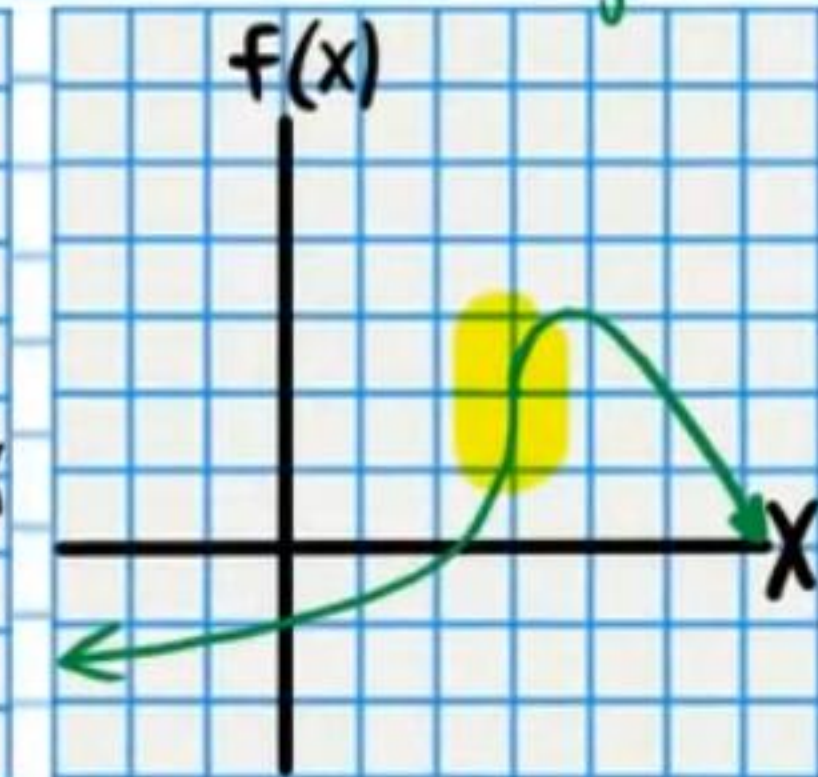


If a function is continuous on an interval, then it is differentiable on this interval, except at x-values where its graph has **sharp turns** or **vertical tangent lines**.

Sharp Turns



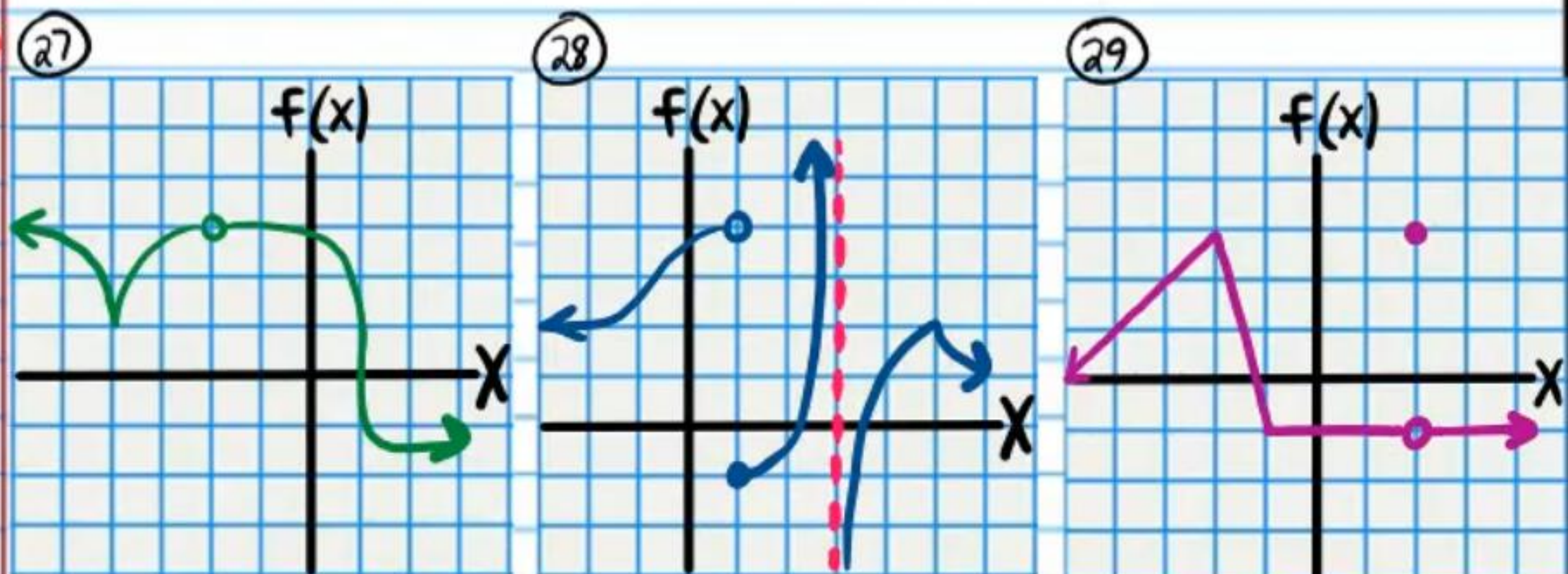
Vertical Tangents



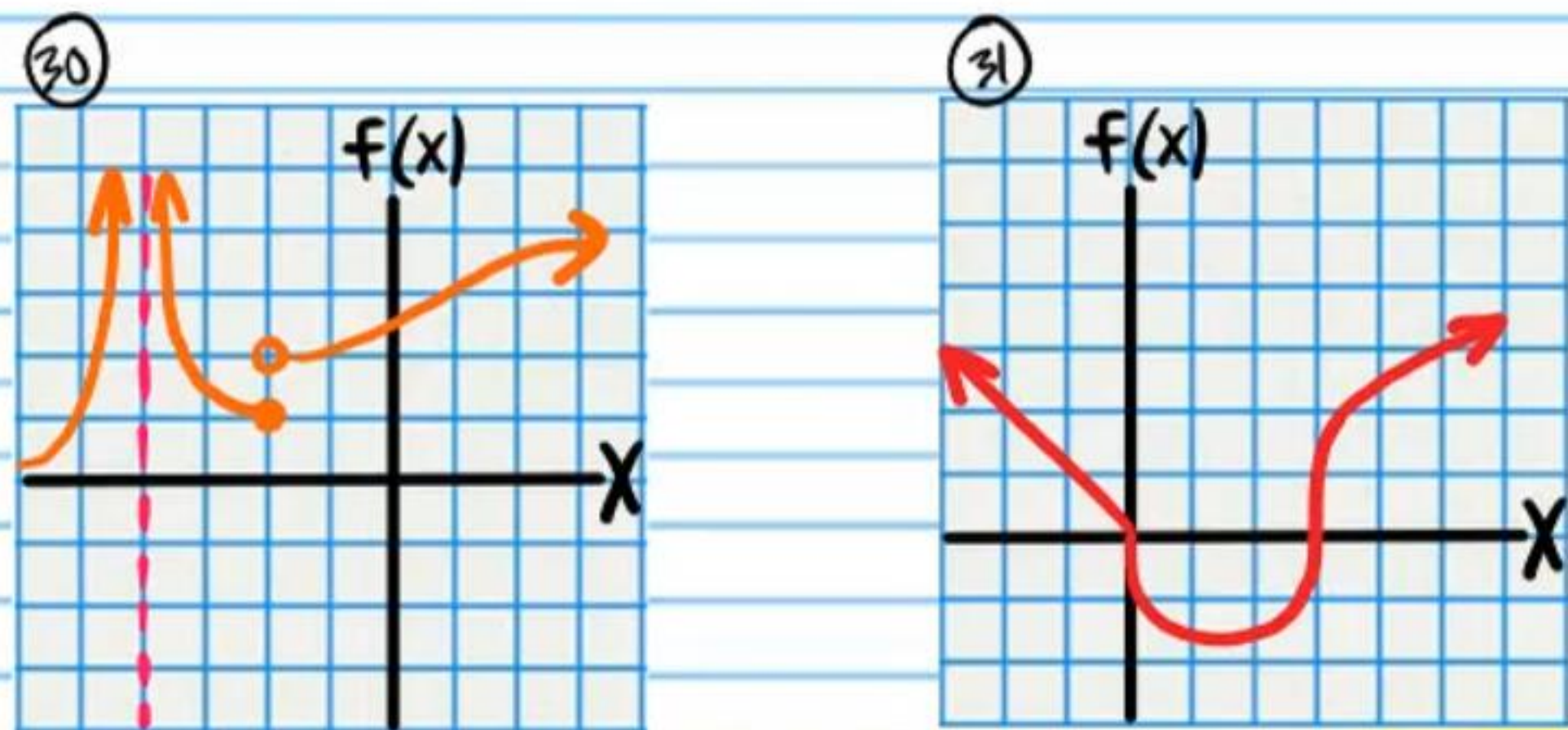
Differentiability Proves Continuity Theorem: If a function is differentiable at $x=a$, then it **must** be continuous at $x=a$.

For problem #'s 27-31,

- a) Write all x -values where there is a discontinuity.
b) Write all x -values where the function is not differentiable.



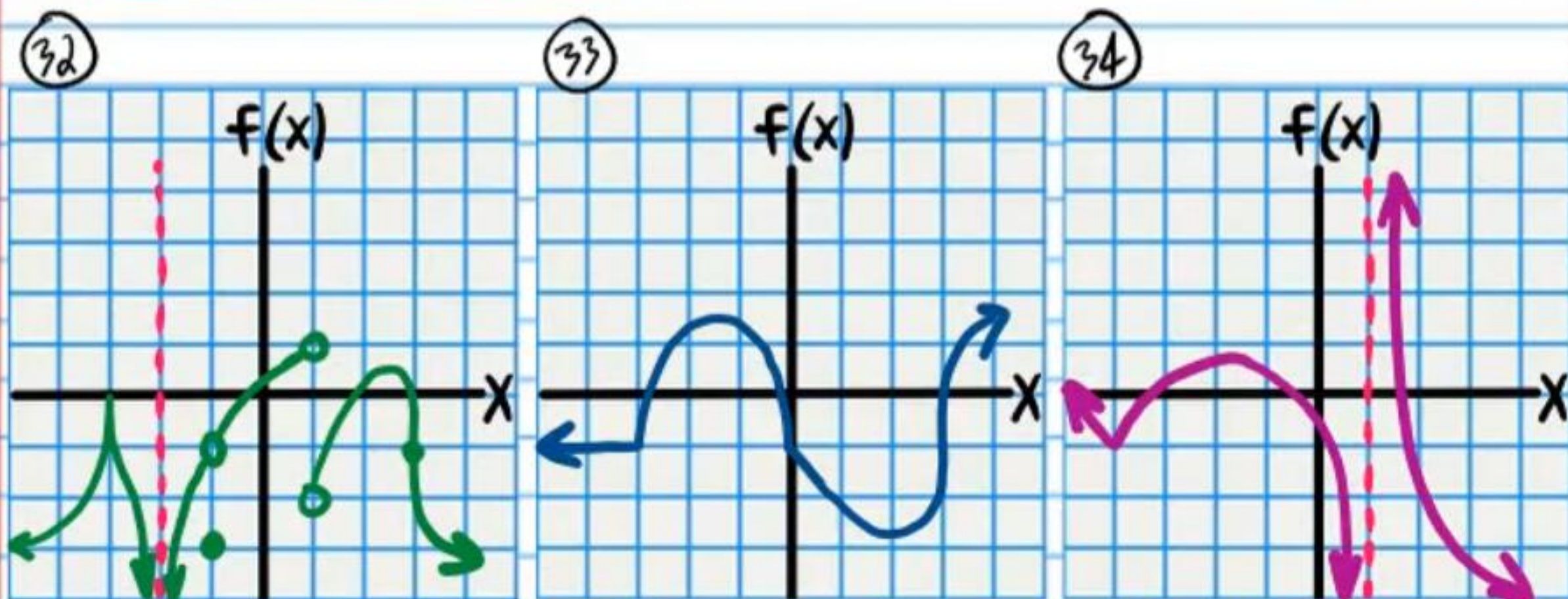
- a) $x = -2$
b) $x = -4, x = -2, x = 1$
- a) $x = 1, x = 3$
b) $x = 1, x = 3, x = 5$
- a) $x = 2$
b) $x = -2, x = -1, x = 2$



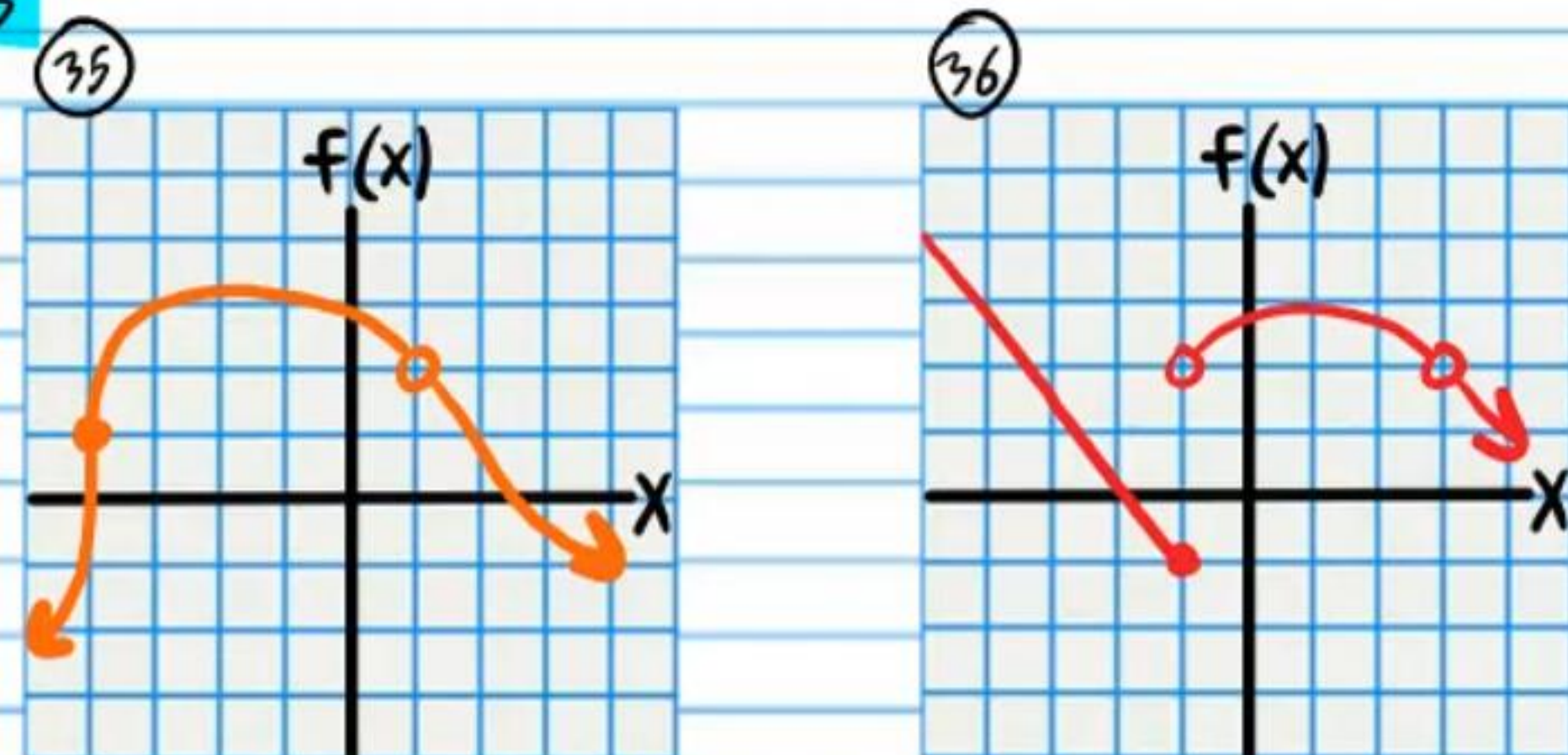
- a) $x = -4, x = -2$
b) $x = -4, x = -2$
- a) None
b) $x = 0, x = 3$

For problem #'s 32-36,

- a) Write all x -values where there is a discontinuity.
b) Write all x -values where the function is not differentiable.



- a) $x = -2, x = -1, x = 1$
b) $x = -3, x = -2, x = -1, x = 1, x = 3$
- a) None
b) $x = -3, x = 0, x = 3$
- a) $x = 1$
b) $x = -4, x = 1$



- a) $x = 1$
b) $x = -4, x = 1$
- a) $x = -1, x = 3$
b) $x = -1, x = 3$

Demonstrate algebraically that each function below is not differentiable at the indicated x -value (a graph of each function is shown at the right).

• (37) $f(x) = 1 - (x+2)^{\frac{2}{3}}$, $x = -2$

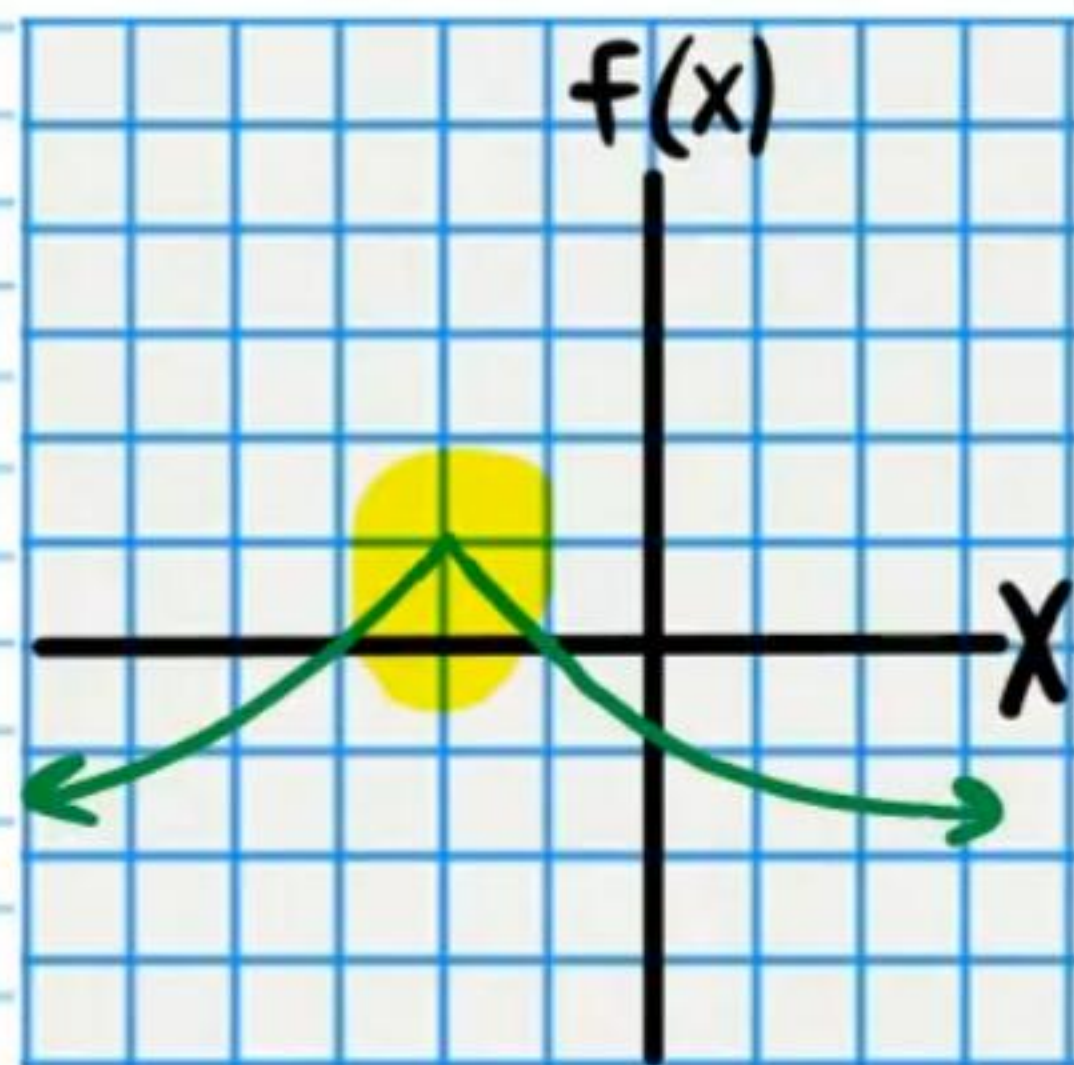
$$f'(x) = 0 - \frac{2}{3}(x+2)^{\frac{2}{3}-1} \frac{d}{dx}(x+2)$$

$$= -\frac{2}{3}(x+2)^{-\frac{1}{3}} (1)$$

$$= -\frac{2}{3(x+2)^{\frac{1}{3}}}$$

$$= -\frac{2}{3\sqrt[3]{x+2}}$$

$f'(-2)$ is undefined.



$$f(-2) = -\frac{2}{3\sqrt[3]{-2+2}}$$

$$= -\frac{2}{3\sqrt[3]{0}}$$

$$= -\frac{2}{3(0)}$$

$$= -\frac{2}{0}$$

(38) $f(x) = 2\sqrt[5]{(x-3)^3} - \sqrt[5]{(x-3)^8} + 2$, $x = 3$

$$f(x) = 2(x-3)^{\frac{3}{5}} - (x-3)^{\frac{8}{5}} + 2$$

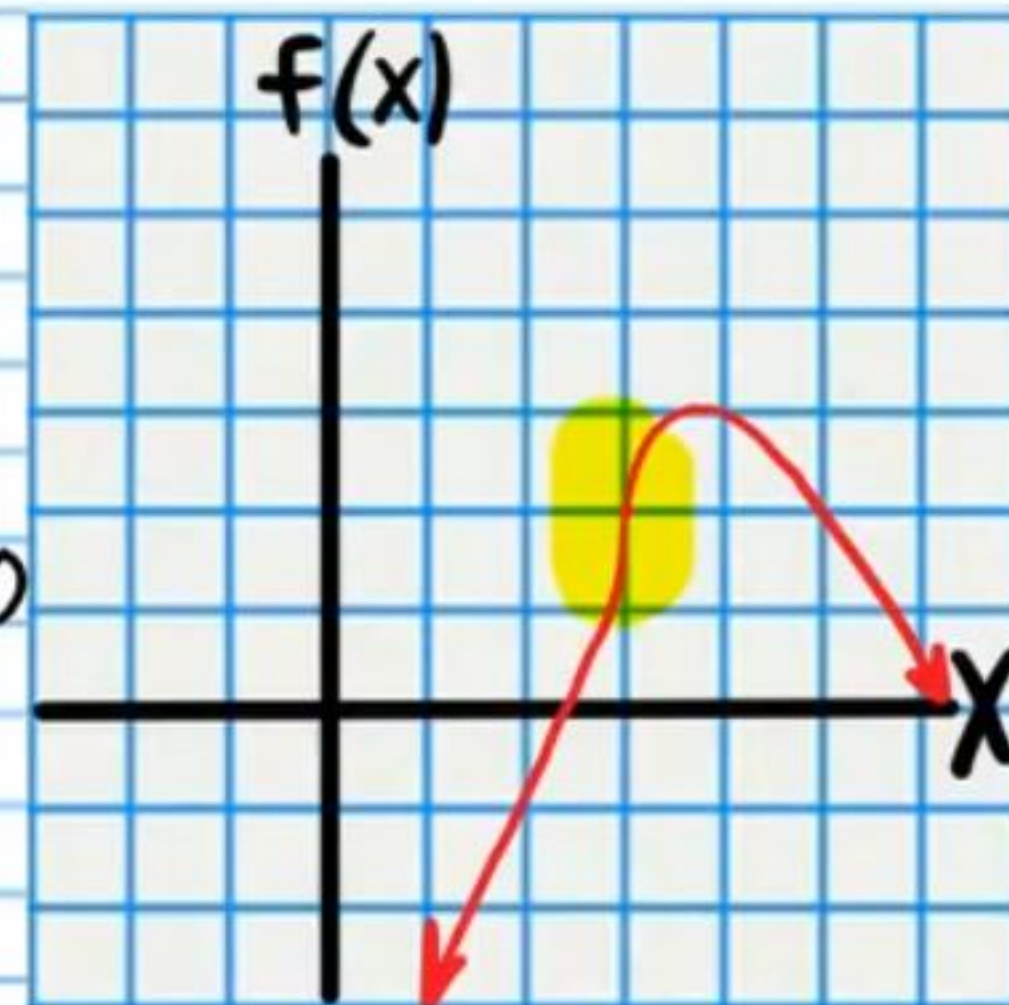
$$f'(x) = 2\left(\frac{3}{5}\right)(x-3)^{\frac{3}{5}-1} \frac{d}{dx}(x-3) - \frac{8}{5}(x-3)^{\frac{8}{5}-1} \frac{d}{dx}(x-3) + 0$$

$$f'(x) = \frac{6}{5}(x-3)^{-\frac{2}{5}}(1) - \frac{8}{5}(x-3)^{\frac{3}{5}}(1)$$

$$f'(x) = \frac{6}{5(x-3)^{\frac{2}{5}}} - \frac{8}{5}\sqrt[5]{(x-3)^3}$$

$$f'(x) = \frac{6}{5\sqrt[5]{(x-3)^2}} - \frac{8}{5}\sqrt[5]{(x-3)^3}$$

$f'(3)$ is undefined.



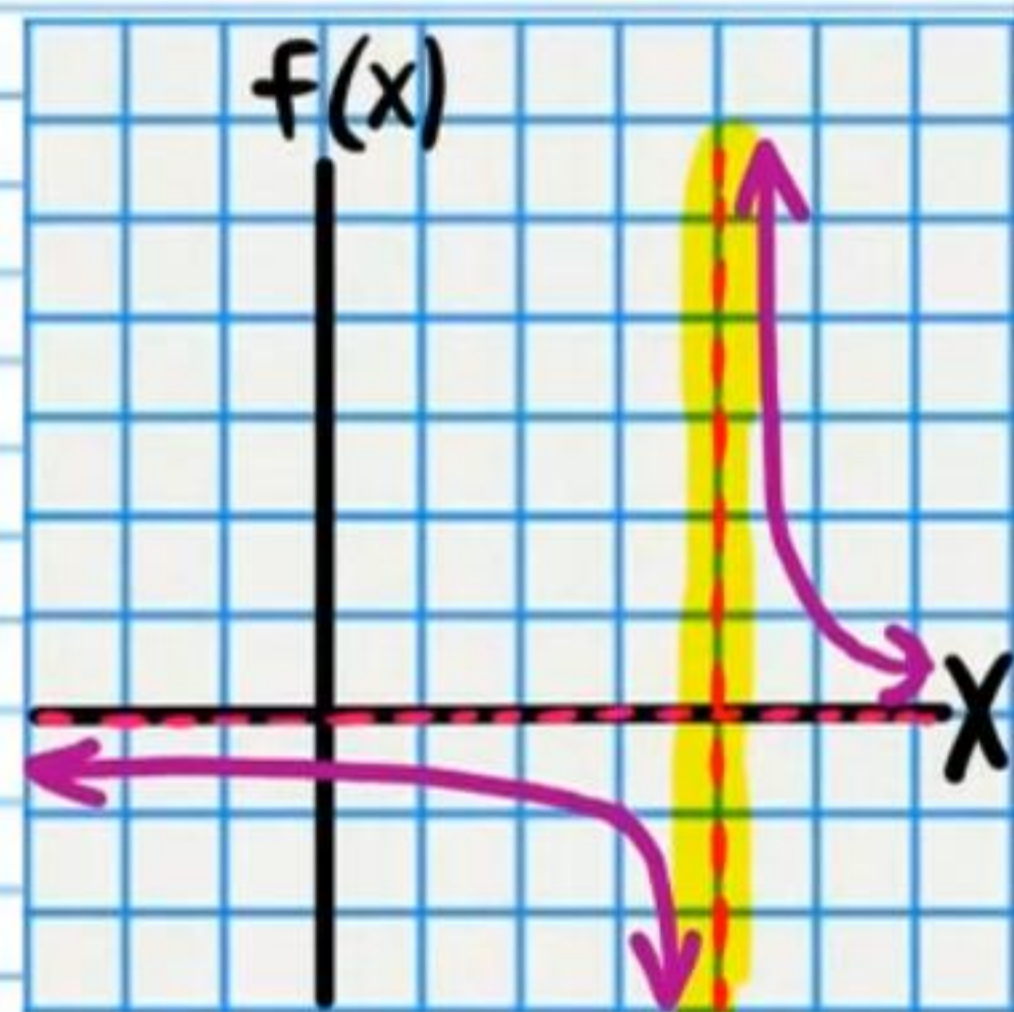
③⑨ $f(x) = \frac{1}{x-4}$, $x=4$

Option 1
 $f'(x) = \frac{\left[\frac{d}{dx}(1) \right] (x-4) - 1 \frac{d}{dx}(x-4)}{(x-4)^2}$

$$f'(x) = \frac{0(x-4) - 1(1)}{(x-4)^2}$$

$$f'(x) = \frac{-1}{(x-4)^2}$$

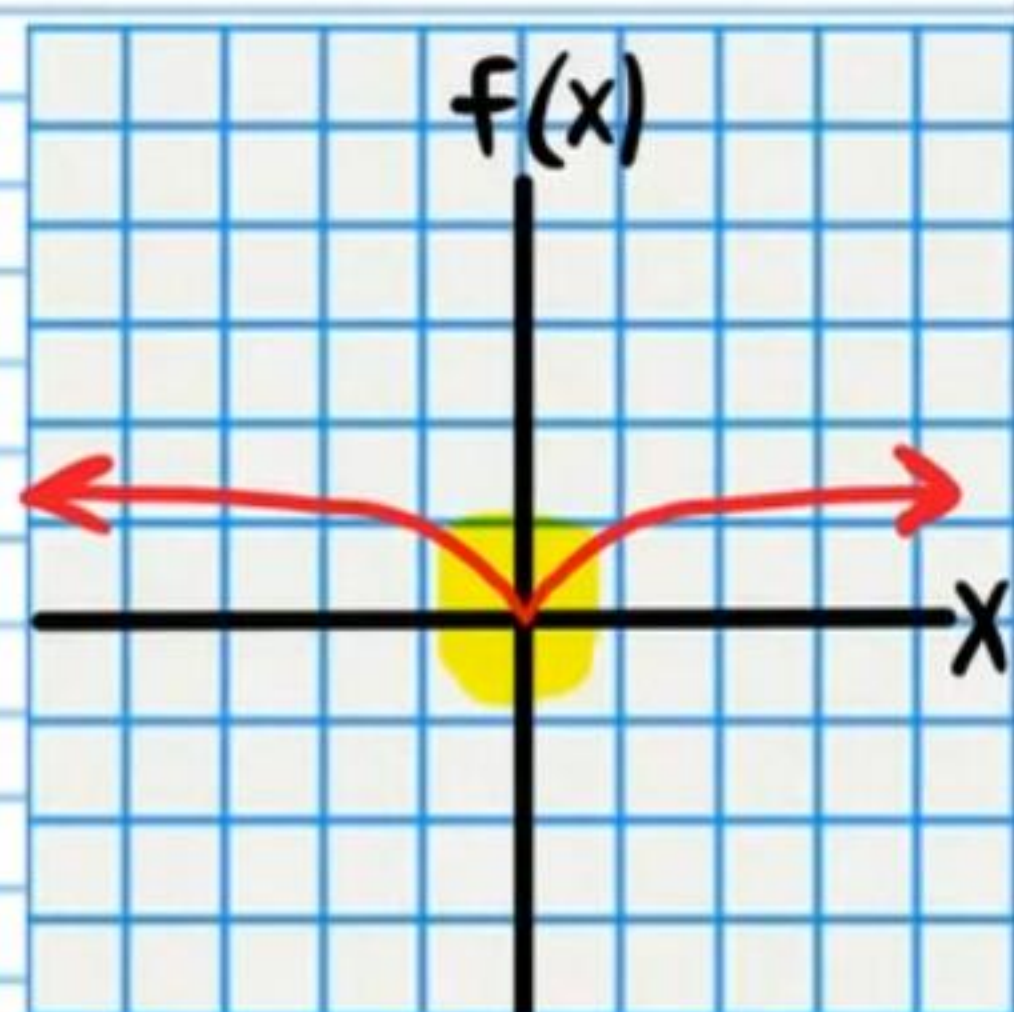
$f'(4)$ is undefined.



Option 2

$$f(x) = \frac{1}{(x-4)^1}$$

$$f(x) = (x-4)^{-1}$$



$f'(0)$ is undefined.

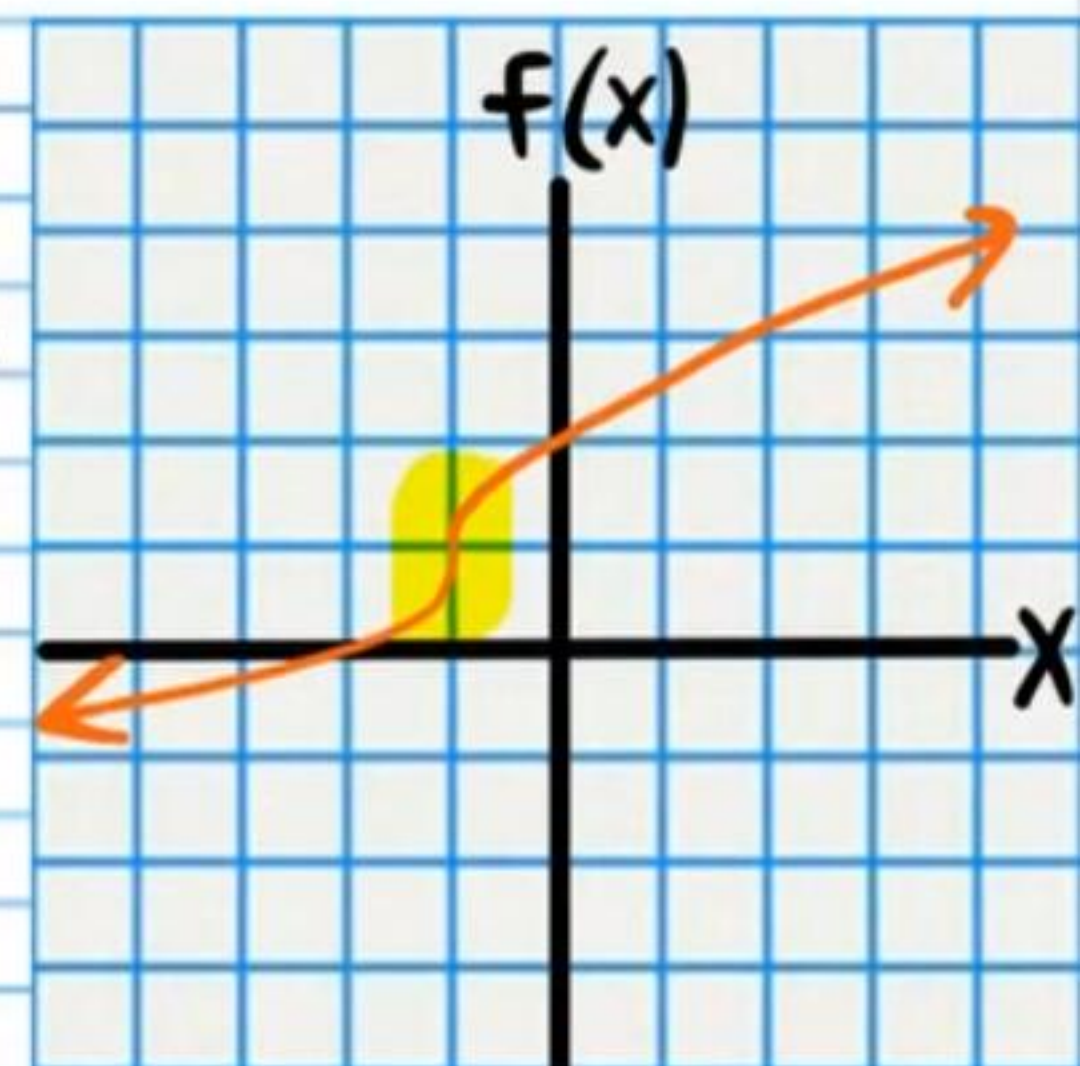
④① $f(x) = (x+1)^{\frac{5}{7}} + 1$, $x=-1$

$$f'(x) = \frac{5}{7} (x+1)^{\frac{5}{7}-1} \frac{d}{dx}(x+1) + 0$$

$$f'(x) = \frac{5}{7} (x+1)^{-\frac{2}{7}} (1)$$

$$f'(x) = \frac{5}{7 \sqrt[7]{(x+1)^2}}$$

$f'(-1)$ is undefined.



④② $f(x) = -\frac{2}{x+1} + 1, x = -1$

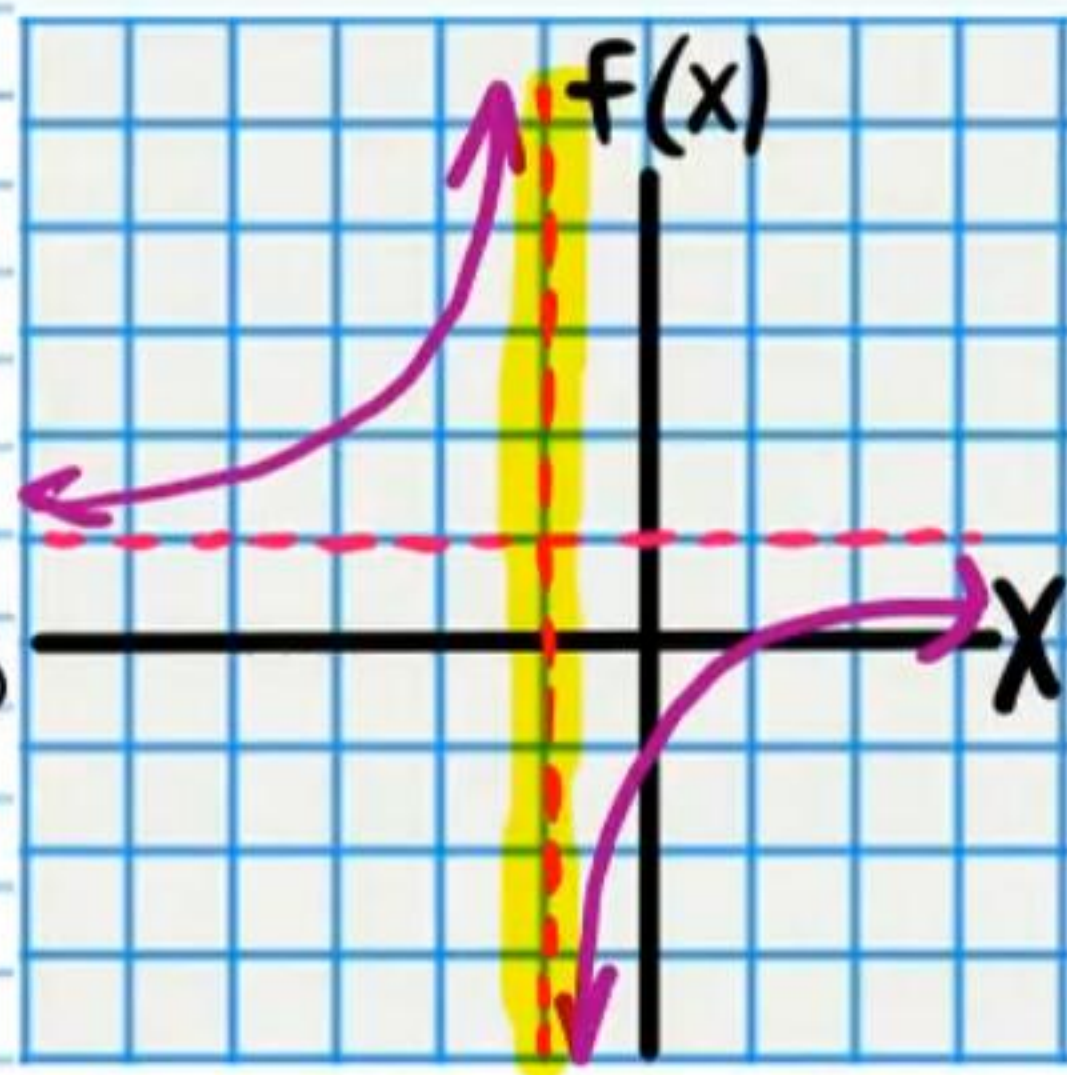
$$f(x) = -2(x+1)^{-1} + 1$$

$$f'(x) = -2(-1)(x+1)^{-1-1} \frac{d}{dx}(x+1) + 0$$

$$f'(x) = 2(x+1)^{-2} (1)$$

$$f'(x) = \frac{2}{(x+1)^2}$$

$f'(-1)$ is undefined.



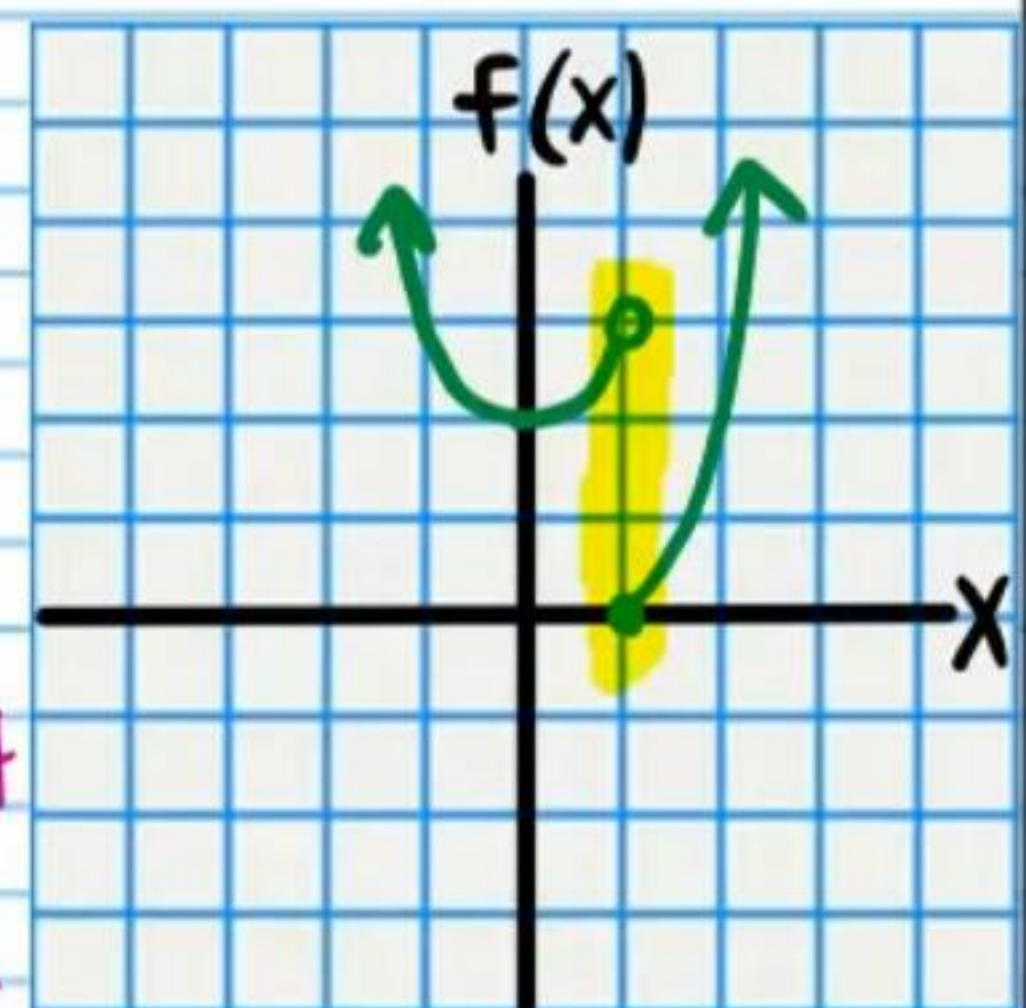
For the following problems, the definition of the derivative is required to demonstrate that each function is not differentiable at the indicated x -value.

④③ $f(x) = \begin{cases} x^2 + 2, & x < 1 \\ x^2 - 1, & x \geq 1 \end{cases} \quad x = 1$

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

In order for this limit to exist, the left and right limits must be equal. Although the right limit exists, the left limit does not, so it is impossible for them to be equal.

$$\lim_{h \rightarrow 0^+} \frac{f(x+h) - f(x)}{h} \stackrel{?}{=} \lim_{h \rightarrow 0^-} \frac{f(x+h) - f(x)}{h}$$

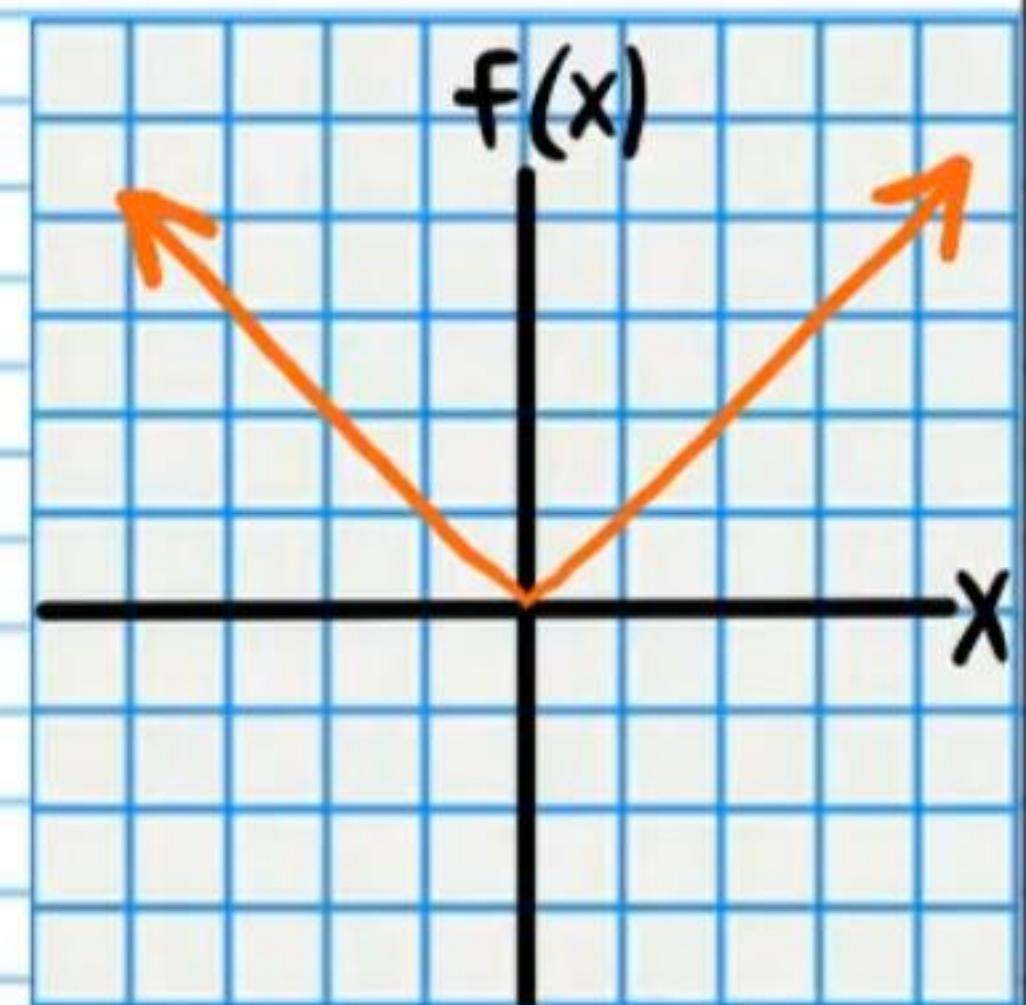


④④ $f(x) = |x| \quad x = 0$

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

In order for this limit to exist, the left and right limits must be equal.

$$\lim_{h \rightarrow 0^+} \frac{f(x+h) - f(x)}{h} \stackrel{?}{=} \lim_{h \rightarrow 0^-} \frac{f(x+h) - f(x)}{h}$$



But upon investigation, we find that they are not.